AIMS Mathematics

Supplementary Material

Safe Active Discrimination of Fractional, Delayed, and Finite Latent Memory in a Strong-Allee Predator–Prey Model

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The main paper was deliberately shortened after external feedback. This supplement preserves the detailed theoretical development, proofs, secondary results, Bayesian extension, and numerical validation that were removed from the main narrative. The main paper contains only the four analytical results directly supporting the experiments.

S1. Certified strong-Allee ecological backbone

The strong-Allee predator–prey backbone in this supplement is inherited from the previously submitted validated study [25]. That paper contains the derivation and certification of the coexistence branch, the Matignon stability boundary, the extinction funnel, global boundedness, and the discretization counter-example. Those results are not re-proved or claimed as new here. We record only the algebraic quantities needed by the new transfer-function, experiment-design, and model-discrimination analysis.

The inherited nonlinear vector field is

$$f_1(x,y) = rx\left(1-\frac{x}{K}\right)\left(\frac{x}{A}-1\right) - \frac{axy}{1+hx}, \tag{S1}$$

$$f_2(x,y) = e\frac{axy}{1+hx} - my, \tag{S2}$$

with the same locked parameterization as Ref. [25],

$$r = \frac{3}{2}, \quad K = 1, \quad a = 1, \quad h = \frac{1}{2}, \quad e = \frac{4}{5}, \quad m = \frac{2}{5}.$$

For later linear input/output calculations we need only

$$x^* = \frac{2}{3}, \qquad y^*(A) = \frac{2(2-3A)}{9A},$$

and

$$J(A) = \begin{pmatrix} \dfrac{7A-2}{8A} & -\dfrac{1}{2} \\ \dfrac{2-3A}{10A} & 0 \end{pmatrix}.$$

The stability screening used throughout the present experiments is likewise inherited from Ref. [25]. In the complex-pair regime, that study established the critical Matignon order

$$\alpha^*(A) = \frac{2}{\pi}\arctan\left(\frac{\sqrt{4D(A)-T(A)^2}}{T(A)}\right),$$

where $T(A) = \operatorname{tr} J(A)$ and $D(A) = \det J(A)$. We use this already-established boundary only to exclude focal unstable cells from discrimination summaries. The present contribution begins with the competing memory laws and their controlled input/output responses, not with re-certification of this ecological backbone.

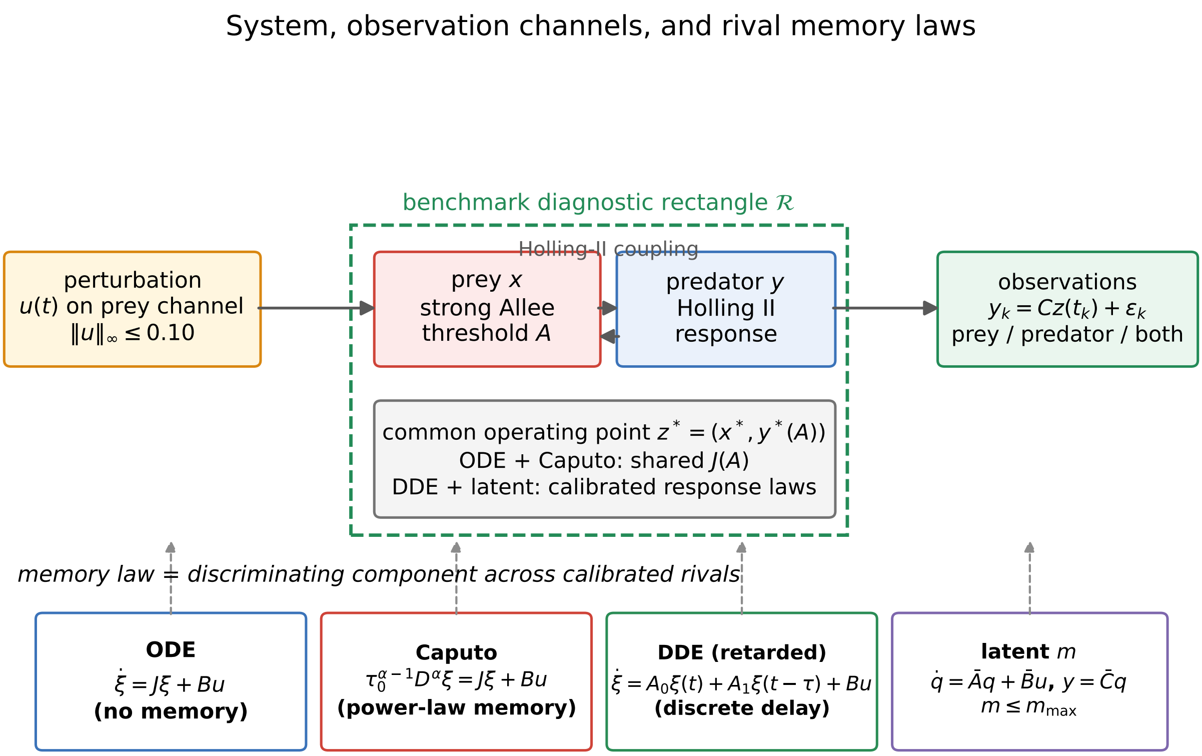

Figure S1. Experimental architecture built on the inherited validated backbone of Ref. [25]. Prey with a strong Allee threshold, predator with a Holling type II response, and coexistence equilibrium z^* receive bounded perturbations $u(t)$ on the prey channel and is observed through prey, predator, or both channels. The ODE and Caputo mechanisms share the ecological Jacobian $J(A)$ exactly; the retarded-DDE and finite-latent alternatives are calibrated at the same operating point and compared through matched input/observation channels and physical units, without assuming identical internal generators. The latent class carries a declared complexity budget $m \le m_{\max}$. The dashed green rectangle is the benchmark diagnostic rectangle; positive invariance requires the strict inward-pointing hypotheses of Section S7 and is not implied by the rectangle definition alone.

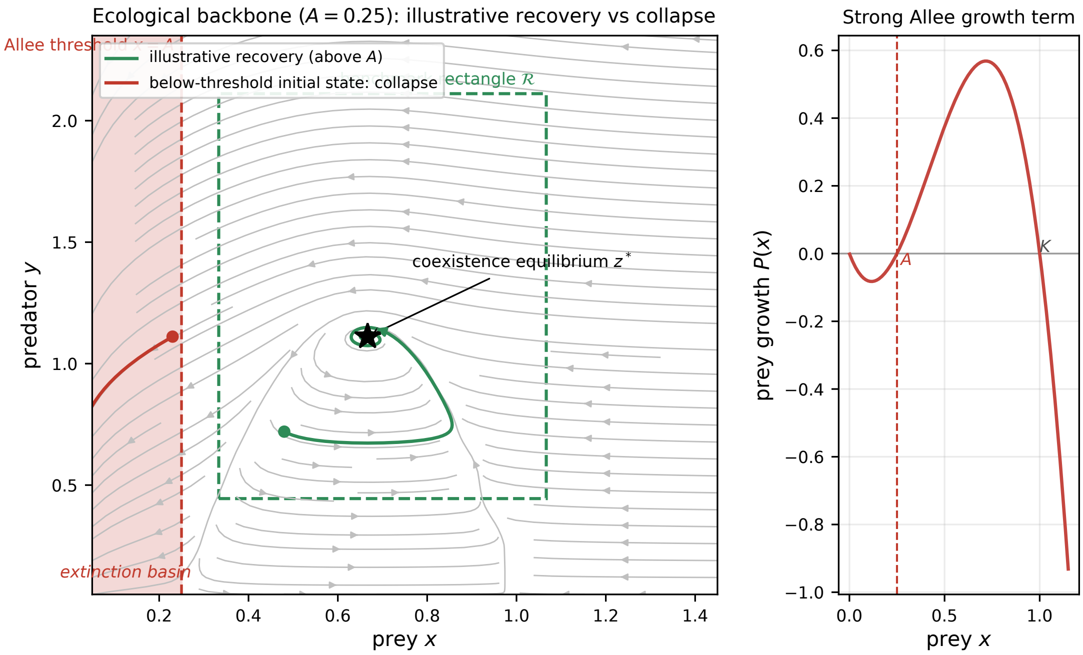

Figure S2. Inherited ecological backbone [25] at the benchmark value $A = 0.25$, shown only to define the safety geometry of the new discrimination experiments. Left: an illustrative phase portrait with the rigorously established Allee threshold $x = A$ and extinction strip, the benchmark diagnostic rectangle $\mathcal{R}$ around z^*, and two numerically integrated trajectories: recovery from above threshold and collapse from below A. The plotted rectangle and trajectories are visual diagnostics, not an invariant-set certificate; invariance requires Theorem S7.4's strict face conditions. Right: the strong-Allee growth term $P(x)$, whose negative branch below A explains why safety cannot be an afterthought.

S2. Dimensionally Consistent Controlled Memory Models

To ensure dimensional consistency when introducing a fractional derivative of order α, we introduce a reference time scale τ_0. The dimensionally consistent commensurate Caputo model for the controlled strong-Allee predator-prey system is

$$\tau_0^{\alpha-1}\,{}_0^C D_t^{\alpha} z(t) = f(z(t);\theta) + Bu(t), \tag{S3}$$

where $z = (x, y)^\top$ is the state vector (prey, predator), f is the ecological vector field, B is the input matrix, and $u(t)$ is the control input. At $\alpha = 1$, equation (S3) reduces exactly to the integer-order ODE.

The ecological vector field f is that of the strong-Allee predator-prey model:

$$f_1(x, y) = rx\left(1 - \frac{x}{K}\right)\left(\frac{x}{A} - 1\right) - \frac{axy}{1 + hx}, \tag{S4}$$

$$f_2(x, y) = e\frac{axy}{1 + hx} - my, \tag{S5}$$

with the locked parameter values $r = 3/2$, $K = 1$, $a = 1$, $h = 1/2$, $e = 4/5$, $m = 2/5$ inherited from the validated baseline.

We consider four candidate dynamic-memory mechanisms around a common experimental operating point. The ODE and Caputo mechanisms share the ecological Jacobian $J(A)$ exactly; the retarded-DDE and finite-state latent alternatives are calibrated rival response laws with the same input/observation channels and physical units, but their internal generators A_0, A_1 and $\bar{A}$ are not literally the same Jacobian:

1. **M_ODE**: integer-order ODE with dynamics $\dot{\xi} = J\xi + Bu$ (memory horizon: none).
2. **M_Caputo**: fractional derivative of order $\alpha \in (0, 1)$ with dynamics $\tau_0^{\alpha-1}\,{}_0^C D_t^{\alpha}\xi = J\xi + Bu$; the order α belongs to this mechanism alone.
3. **M_DDE**: retarded delay differential equation with finite state dimension and a single discrete delay τ, $\dot{\xi} = A_0\xi(t) + A_1\xi(t - \tau) + Bu$.

4. **M_latent,m**: finite-dimensional linear time-invariant latent state model of dimension m with dynamics $\dot{q} = \bar{A}q + \bar{B}u$, $y = \bar{C}q$.

Only **M_Caputo** carries a fractional order; the ODE, DDE, and latent mechanisms are integer-order. They are compared through matched input and observation channels and a common observation-noise model. The fractional transfer is $G_\alpha(s) = C(q_\alpha(s)I - J)^{-1}B$, whereas the DDE and latent classes use their own declared generators. Figure S3 summarizes this distinction.

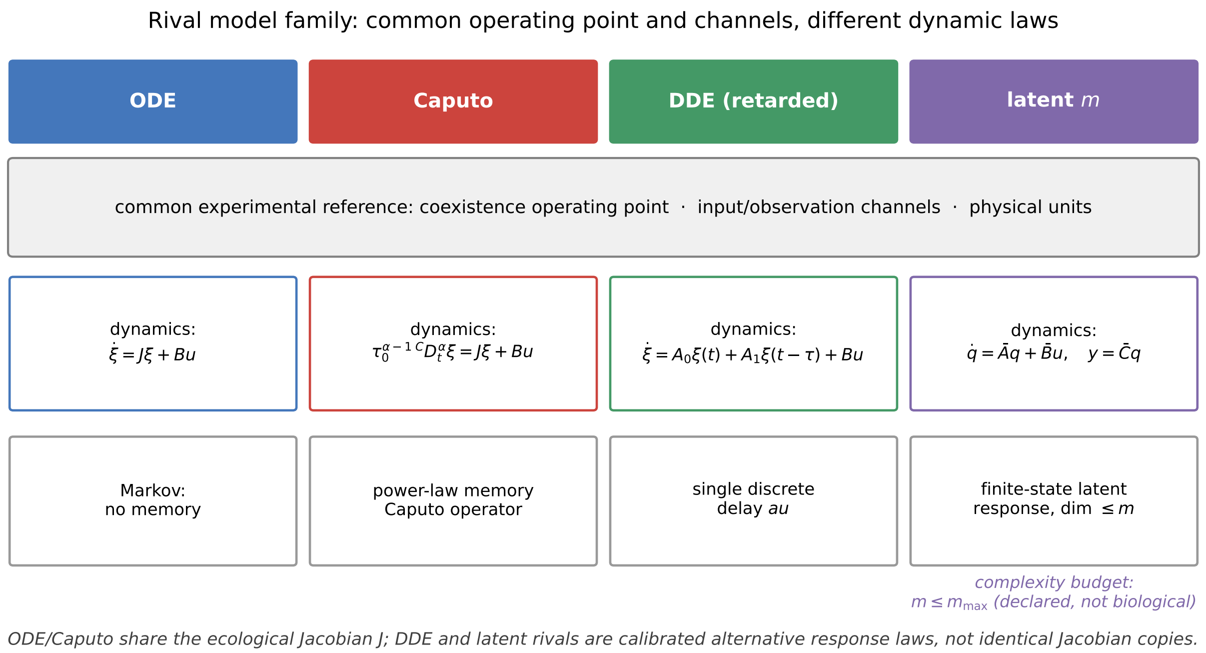

Figure S3. The rival model family. ODE and Caputo dynamics share the ecological Jacobian exactly; retarded-delay and finite-state latent rivals are alternative response laws calibrated at the same experimental operating point and compared through matched input/observation channels and units. The latent class is bounded by a declared complexity budget $m \leq m_{\max}$, not by a biological law.

The dimensional consistency enforced by τ_0 [12, 19, 31] ensures that the fractional order α does not spuriously alter the units of the ecological rates, allowing a fair comparison between the memory hypotheses.

S3. Exact Linearization, Transfer Functions, and Channels

Starting from the equilibrium and Jacobian inherited from Ref. [25], we now derive the input/output objects that are specific to the present discrimination problem: channel transfer functions, controllability/observability statements, and their high-frequency signatures [8, 22, 27].

The coexistence Jacobian can be written as

$$J = \begin{pmatrix} T & -c \\ d & 0 \end{pmatrix}, \qquad c = \frac{1}{2}, \qquad d = \frac{2-3A}{10A} > 0, \tag{S6}$$

where the trace $T(A) = \frac{7A-2}{8A}$ and determinant $D(A) = \frac{2-3A}{20A}$ are inherited backbone quantities from Ref. [25].

The common fractional order changes the temporal symbol $s \mapsto q_\alpha(s) = \tau_0^{\alpha-1}s^\alpha$ but does not alter the algebraic controllability/observability of the pair (J, B, C) [17].

For prey intervention $B_x = e_1$, the controllability matrix $\mathcal{C}_x = [B_x, JB_x]$ has determinant $d > 0$. For predator intervention $B_y = e_2$, $\mathcal{C}_y = [B_y, JB_y]$ has determinant $c > 0$. Thus either prey or predator perturbation alone is sufficient to excite both linearized modes.

For prey-only observation $C_x = e_1^\top$, the observability matrix $\mathcal{O}_x = [C_x; C_xJ]$ has determinant $-c \neq 0$. For predator-only observation $C_y = e_2^\top$, $\mathcal{O}_y = [C_y; C_yJ]$ has determinant $-d \neq 0$. Hence either species alone observes both linearized states; the statement that both species must be observed for structural observability is false.

The four scalar transfer functions are

$$G_{x \leftarrow x}(s) = \frac{q}{\Delta_\alpha(s)}, \tag{S7}$$

$$G_{y \leftarrow x}(s) = \frac{d}{\Delta_\alpha(s)}, \tag{S8}$$

$$G_{x \leftarrow y}(s) = \frac{-c}{\Delta_\alpha(s)}, \tag{S9}$$

$$G_{y \leftarrow y}(s) = \frac{q - T}{\Delta_\alpha(s)}, \tag{S10}$$

where $\Delta_\alpha(s) = q^2 - Tq + D$ and $q = q_\alpha(s)$.

For a general channel $G = C(qI - J)^{-1}B$, the high-frequency expansion is

$$G(s) = \frac{CB}{q} + \frac{CJB}{q^2} + O(|q|^{-3}). \tag{S11}$$

If $CB \neq 0$, the leading decay is $s^{-\alpha}$. If $CB = 0$, the leading term may be $s^{-2\alpha}$ or lower.

For the biological channels:

- Prey intervention + prey observation: $CB = 1$,
- Predator intervention + predator observation: $CB = 1$,
- Cross-species channels: $CB = 0$.

Therefore a collocated perturb-and-measure protocol gives the cleanest high-frequency signature of α. Cross-species measurements remain useful for interaction parameters; their own high-frequency phase differs from the collocated signature (S15) because $CB = 0$ there (Remark S4.6 and Section S6).

Latent environmental states are observable exactly when $C(J + \lambda I)^{-1} g \neq 0$ (Popov–Belevitch–Hautus latent-mode condition). This condition precisely determines when an environmental covariate is needed: not always, but whenever the latent coupling lies in an output-invisible direction or is nearly invisible in the practical noise metric.

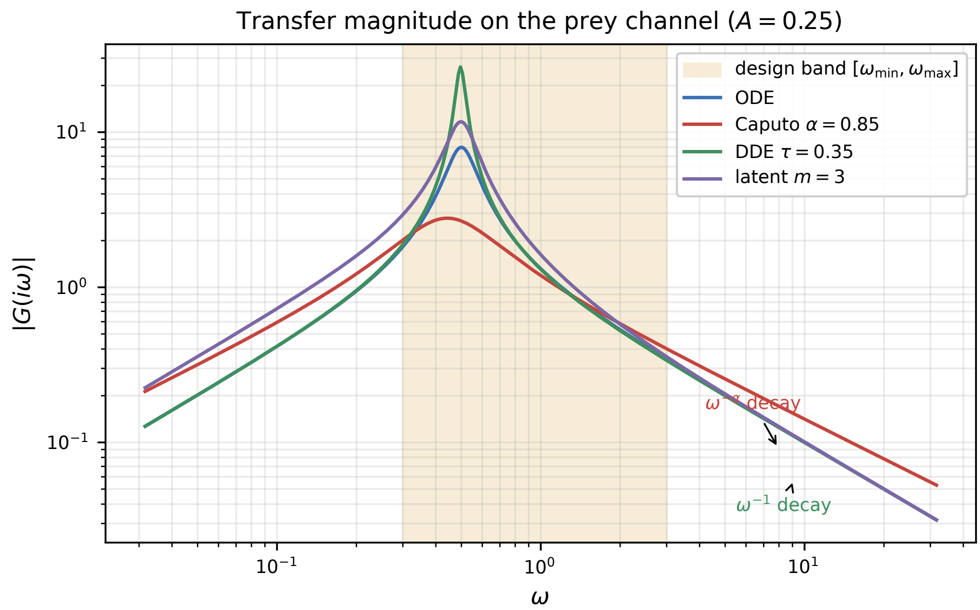

Figure S4. Transfer magnitude $|G(i\omega)|$ on the prey channel for the four memory mechanisms at $A = 0.25$. The shaded region is the design band where the optimal inputs of Section S6 place their energy; inside it the Caputo curve already separates from its rivals, with the $\omega^{-\alpha}$ (Caputo) versus ω^{-1} (DDE) asymptotics taking over at higher frequency.

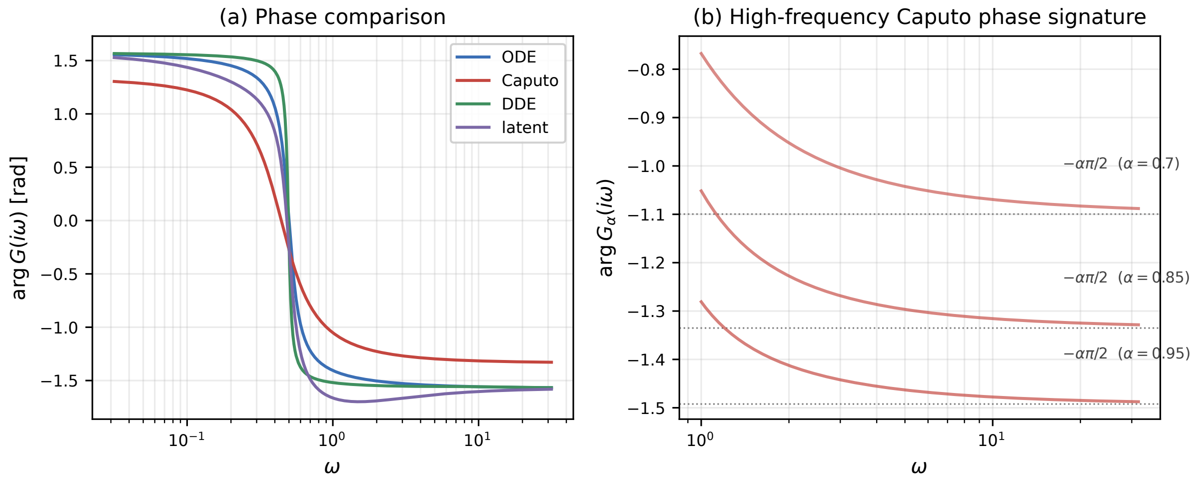

Figure S5. Phase signatures of structural separation. Panel (a): full phase comparison on the prey channel — the Caputo mechanism separates from the integer-order, delay, and latent alternatives. Panel (b): the scale-free high-frequency signature $\arg G_\alpha(i\omega) \to -\alpha\pi/2$ of Equation (S15) for three fractional orders, converging to the dotted asymptotes. This is a collocated-channel property: cross channels ($CB = 0$) carry a different asymptotic phase.

S4. Structural Separation from Finite Latent and Delay Models

The central theoretical result of this paper is that under the direct input–output channel condition $CB \neq 0$, a Caputo fractional-order model is not equivalent to any finite-dimensional latent-state model nor to any finite discrete-delay model [12, 14, 17, 19, 29, 31]. This structural non-equivalence is what makes the discrimination problem well-posed and what the optimal experimental design exploits.

S4.1. The fractional signature in the high-frequency asymptote

Consider the linearized Caputo transfer function

$$G_\alpha(s) = C(q_\alpha(s)I - J)^{-1}B, \qquad q_\alpha(s) = \tau_0^{-1}(s\tau_0)^\alpha, \tag{S12}$$

with $0 < \alpha < 1$ and the principal branch of the fractional power.

For $|q_\alpha(s)| > \|J\|$, the Neumann expansion gives

$$(q_\alpha I - J)^{-1} = q_\alpha^{-1}(I + Jq_\alpha^{-1} + O(|q_\alpha|^{-2})). \tag{S13}$$

Hence

$$G_\alpha(s) = (CB)\tau_0^{1-\alpha}s^{-\alpha} + O(|s|^{-2\alpha}). \tag{S14}$$

The leading term is $s^{-\alpha}$ — a non-integer power. This is the essential feature that distinguishes it from all finite-dimensional models.

S4.2. Theorem T4: Caputo transfer function is not rational

Theorem S4.1 (Structural non-equivalence with latent ODE models)**.** *There is no finite-dimensional integer-order state-space model with transfer function*

$$G_L(s) = D + \bar{C}(sI - \bar{A})^{-1}\bar{B}$$

that equals $G_\alpha(s)$ on a nonempty open subset of their common analytic domain.

Proof. A finite-dimensional integer-order transfer function is rational. If it is strictly proper, its Laurent expansion at infinity contains only integer powers s^{-r} with $r \in \mathbb{N}$. If it has a direct feedthrough, it tends to a nonzero constant. Neither can equal the non-integer leading power in (S14) because $CB \neq 0$ and $0 < \alpha < 1$. Equality on an open set would imply equality by analytic continuation, contradiction. Therefore no finite-dimensional rational transfer function can reproduce the fractional channel. □

Theorem T4 is an exact statement: in the absence of model uncertainty and measurement noise, full-bandwidth input–output data would structurally distinguish a Caputo model from every finite-dimensional ODE latent model. However, it does *not* guarantee practical distinguishability on a finite bandwidth, finite horizon, or under noise. That is why the remaining sections develop explicit quantitative bounds and optimal input design.

S4.3. Theorem T5 – Caputo is not equivalent to a finite retarded delay model

Theorem S4.2 (Structural non-equivalence with retarded single-delay models)**.** *Let*

$$G_\tau(s) = C_\tau(sI - A_0 - A_1 e^{-s\tau})^{-1} B_\tau$$

be the transfer function of a retarded delay system with finite state dimension, a single discrete delay τ, no direct feedthrough, and no derivative output. Then along $s = i\omega$,

$$G_\tau(i\omega) = O(\omega^{-1}), \qquad \omega \to \infty,$$

whereas $G_\alpha(i\omega) = (CB)\tau_0^{1-\alpha}(i\omega)^{-\alpha} + O(\omega^{-2\alpha})$. Since $\alpha \neq 1$, the two transfer functions are not identical at high frequencies and therefore cannot be identical as analytic input–output maps.

Proof. Because $|e^{-i\omega\tau}| = 1$, the matrix $A_0 + A_1 e^{-i\omega\tau}$ remains bounded for all ω. Factor $i\omega I$ from the delay resolvent; a Neumann expansion yields $G_\tau(i\omega) = O(\omega^{-1})$. The fractional asymptotic follows from Theorem T4's expansion (S14). For $0 < \alpha < 1$, the high-frequency decay rates differ, and no model of the stated retarded single-delay class can reproduce the $s^{-\alpha}$ signature. □

Remark S4.3 (Scope of Theorem S4.2). A delay equation is infinite-dimensional as a dynamical system; what is finite in Theorem S4.2 is the state dimension of the underlying ODE skeleton. The theorem covers strictly proper retarded systems with finitely many discrete delays (the proof extends term-by-term to any finite delay set) and no direct feedthrough. It does *not* cover neutral delay equations, distributed delays, derivative outputs, or direct feedthrough; those classes require separate arguments. Throughout the paper, "DDE model" means this finite-state retarded class.

Together, Theorem T4 and Theorem T5 establish a firm boundary: the fractional memory effect, characterized by the power-law tail $s^{-\alpha}$, cannot be absorbed into either a finite-dimensional integer-order latent description or a retarded finite-state delay dynamics of the class above. This structural gap is what the experimental design must amplify.

S4.4. Quantitative Neumann remainder (Theorem T6)

For frequency-domain analysis, we need a quantitative bound on how closely the leading term approximates the full transfer function.

Theorem S4.4 (Quantitative high-frequency remainder)**.** *If $|q_\alpha(s)| \geq 2\|J\|$, then*

$$\left\| G_\alpha(s) - \frac{CB}{q_\alpha(s)} \right\| \leq \frac{2\|C\|\,\|J\|\,\|B\|}{|q_\alpha(s)|^2}.$$

Proof. Write $(qI - J)^{-1} = q^{-1}(I - J/q)^{-1}$. Under $\|J/q\| \leq 1/2$, the norm of $(I - J/q)^{-1} - I$ is bounded by $2\|J/q\|$. Multiplying by C (left) and B (right) and by $|q|^{-1}$ yields the claim. □

S4.5. High-frequency phase signature

When $CB > 0$ is real, evaluating (S14) along the imaginary axis gives the clean diagnostic:

$$\boxed{\arg G_\alpha(i\omega) \to -\frac{\alpha\pi}{2}, \qquad \omega \to \infty.} \tag{S15}$$

A direct integer-order channel $G_0(s) = CB/s$ has asymptotic phase $-\pi/2$. The phase gap $(1-\alpha)\pi/2$ is a scale-free signature of α that can be measured after gain normalization, subject to actuator bandwidth and measurement noise constraints.

For a cross channel with $CB = 0$ the leading term of the expansion is instead proportional to $s^{-2\alpha}$ (or lower), so its asymptotic phase is generally different from $-\alpha\pi/2$. The signature (S15) and the gap $(1-\alpha)\pi/2$ are therefore properties of *collocated* channels and must not be attributed to cross-species measurements; cross channels serve to estimate interaction parameters and their own (different) high-frequency phase, as detailed in Section S3.

S4.6. Discrimination under finite models

The structural separation results guarantee that for any finite set of fixed, structurally distinct transfer functions $\{G_1, \ldots, G_M\}$, there exists a separating frequency.

Theorem S4.5 (Generic separating frequency on a compact band)**.** *Let $G_1, \ldots, G_M$ be scalar functions analytic on a connected open set $D \subset \mathbb{C}$, and let $K = \{i\omega : \omega \in [\omega_{\min}, \omega_{\max}]\} \subset D$ be a compact segment of the imaginary axis containing no poles and no branch point. Assume $G_i - G_j \not\equiv 0$ on D for every pair $i \neq j$. Then each zero set*

$$Z_{ij} = \{\omega \in [\omega_{\min}, \omega_{\max}] : G_i(i\omega) = G_j(i\omega)\}$$

is finite, hence $\bigcup_{i<j} Z_{ij}$ is finite, and every frequency outside that union separates all fixed model pairs simultaneously.

Proof. For each pair, $G_i - G_j$ is nonzero and analytic on D, so its zeros are isolated. An infinite subset of the compact set K would have an accumulation point in K, contradicting the identity theorem. Hence each Z_{ij} is finite; a finite union over model pairs is finite. □

Remark S4.6 (Why the band must be compact). On an unbounded frequency interval the zeros of a nonzero analytic difference can be countably infinite, accumulating only at infinity; the exceptional set is then infinite and the conclusion fails. Theorem S4.5 is therefore stated on a compact band K inside a common analytic domain. The statement also concerns a finite collection of *fixed* transfer functions and provides no uniform separation over continuously parameterized rival classes. Panel (a) of Figure S8 shows the separation numerically on the design band.

The practical interpretation is:

1. **Exact separation is structurally guaranteed** for fixed models — at almost all frequencies.
2. **Robust separation is the true challenge:** the inter-model response difference can be orders of magnitude smaller than measurement noise.
3. **Adaptive models can close the gap:** a parameter-tunable latent or delay model can match the fractional response at one or several frequencies by tuning its internal parameters. This is why optimal multiscale excitation that spans a broad frequency band is necessary — not merely a single "informative" frequency.

These observations motivate Sections S5 and S6, where we develop the quantitative finite-horizon tools and optimal input construction needed to convert structural separation into a practical discrimination experiment.

S5. Finite-Horizon Exponential Approximation and Impossibility Barrier

Section S4 established that a Caputo fractional transfer function is structurally distinct from any finite-dimensional integer-order or finite-delay model. However, structural non-equivalence in the infinite-bandwidth, zero-noise limit does not guarantee practical separability in a finite experiment. This section develops the quantitative tools needed to bridge that gap: an exact representation of fractional memory as a continuum of exponentials, a constructive finite-sum approximation with explicit L^1 error bounds [4, 5, 16, 24, 34], and a fundamental impossibility result that either justifies or forces our finite-latent-dimension restriction.

All formulas use nondimensional time; restore physical units with t/τ_0.

S5.1. Exact positive diffusive representation

The Caputo fractional integral kernel

$$k_\alpha(t) = \frac{t^{\alpha-1}}{\Gamma(\alpha)}$$

admits an exact integral representation as a weighted continuum of exponential relaxation modes.

Theorem S5.1 (Exact positive diffusive representation)**.** *For $0 < \alpha < 1$,*

$$\boxed{k_\alpha(t) = \frac{\sin(\pi\alpha)}{\pi} \int_0^\infty \lambda^{-\alpha} e^{-\lambda t}\, d\lambda.} \tag{1}$$

The derivative-memory kernel also satisfies

$$\boxed{\frac{t^{-\alpha}}{\Gamma(1-\alpha)} = \frac{\sin(\pi\alpha)}{\pi} \int_0^\infty \lambda^{\alpha-1} e^{-\lambda t}\, d\lambda.} \tag{2}$$

Proof. Apply the integral identity

$$\int_0^\infty \lambda^{\nu-1} e^{-\lambda t} d\lambda = \Gamma(\nu) t^{-\nu}$$

and Euler's reflection formula $\Gamma(\alpha)\Gamma(1-\alpha) = \pi/\sin(\pi\alpha)$. Set $\nu = 1-\alpha$ for (1) and $\nu = \alpha$ for (2). □

The crucial consequence is that *fractional memory is exactly an infinite continuum of exponential relaxation modes* [12, 19, 30, 31]. Any finite-dimensional latent ODE model is precisely a quadrature approximation to this continuum. This observation makes the "fractional versus latent" question rigorous: it asks how many quadrature nodes are needed before the approximation becomes indistinguishable from the true fractional dynamics within the noise budget of the experiment.

S5.2. Uniform approximation at the singular endpoint is impossible

Theorem S5.2 (No uniform approximation at zero)**.** *No finite exponential sum*

$$k_m(t) = \sum_{j=1}^{m} c_j e^{-\lambda_j t}$$

with finite coefficients can approximate k_α uniformly on $[0,T]$ when $0 < \alpha < 1$.

Proof. Every finite sum has a finite value $k_m(0) = \sum_j c_j$. But $\lim_{t\downarrow 0} k_\alpha(t) = +\infty$. Hence $\sup_{0\le t\le T} |k_\alpha(t) - k_m(t)| = \infty$. □

This result corrects any claim of uniform absolute-error approximation on an interval containing the origin. The fractional kernel is singular at $t = 0$, and no finite sum of exponentials can reproduce this singularity. However, L^p approximation in the integral sense *is* possible, and that is what the experimental design requires: the output signal is a convolution of the kernel with the input, and convolution integrates out the singularity.

S5.3. Constructive finite L1 approximation

Theorem S5.3 (Constructive L^1 approximation)**.** *Let $c_\alpha = \sin(\pi\alpha)/\pi$. Choose $0 < \ell < L$, divide $[\ell, L]$ into m equal intervals of width $h = (L-\ell)/m$, and use left nodes $\lambda_j = \ell + (j-1)h$. Define*

$$k_m(t) = c_\alpha h \sum_{j=1}^{m} \lambda_j^{-\alpha} e^{-\lambda_j t}. \tag{3}$$

Then

$$\boxed{\|k_\alpha - k_m\|_{L^1(0,T)} \le c_\alpha \left[\frac{T\ell^{1-\alpha}}{1-\alpha} + \frac{L^{-\alpha}}{\alpha} + Th\ell^{-\alpha}\right].} \tag{4}$$

Proof idea. Split the integral into head $(0,\ell)$, body $[\ell, L]$, and tail (L,∞) of the diffusive representation; the head and tail are controlled by the explicit window choices and the body by composite quadrature on a geometric grid. The complete proof, together with the explicit parameter choices achieving a prescribed tolerance ε, is given in the Supplementary Material (Section S2). □

S5.4. Output approximation via convolution

Theorem S5.4 (Output approximation)**.** *For any input $u \in L^2(0,T)$,*

$$\boxed{\|(k_\alpha - k_m) * u\|_{L^2(0,T)} \le \|k_\alpha - k_m\|_{L^1(0,T)} \, \|u\|_{L^2(0,T)}.} \tag{5}$$

Proof. Young's convolution inequality. □

Theorem S5.4 is the bridge that converts the kernel approximation error into an output-signal bound: if the L^1 kernel gap can be driven below $\eta/\|u\|_2$, the output signal difference falls below η.

S5.5. Theorem T10 – the finite-horizon impossibility barrier

Theorem S5.5 (Finite-horizon no-free-lunch)**.** *Fix a horizon T, an input $u \in L^2(0,T)$, and any tolerance $\eta > 0$. There exists a finite positive exponential mixture k_m such that*

$$\|(k_\alpha - k_m) * u\|_{L^2(0,T)} < \eta.$$

Proof. By Theorem S5.3, choose m so that $\|k_\alpha - k_m\|_1 < \eta/\|u\|_2$ when $u \neq 0$. Apply Theorem S5.4. The zero-input case is trivial. □

Corollary S5.6 (Unrestricted latent dimension destroys robust discrimination (linearized, fixed input))**.** *In the linear convolution experiment with a fixed input $u \in L^2(0,T)$ and a fixed noise floor, if the alternative latent-state model class allows arbitrary dimension m, then there exists a latent model whose output lies below the noise floor. Therefore any maximin optimal design against an unrestricted latent class achieves* zero robust separation *in this linearized fixed-input sense.*

Remark S5.7 (Exact scope of Corollary S5.6). The corollary is a *kernel-level* statement: it follows from Theorems S5.3–S5.4, which bound the convolution of the fractional kernel against a fixed input. By itself it does *not* prove that the full nonlinear Caputo predator–prey input–output map can be approximated below any noise floor. Extending the barrier to the nonlinear system requires writing the system in Volterra form $x(t) = x_0 + (k_\alpha * F(x,u))(t)$, a common invariant region on which F is Lipschitz (constant L) and bounded (bound M), and a continuous-dependence estimate propagating kernel error into state error; a sufficient version is

$$\|x - x_m\|_{L^\infty(0,T)} \leq \frac{M\,\|k_\alpha - k_m\|_{L^1(0,T)}}{1 - L\,\|k_m\|_{L^1(0,T)}}, \qquad L\,\|k_m\|_{L^1(0,T)} < 1,$$

with more general horizons requiring a Volterra resolvent or fractional Grönwall estimate [38]. Until such a state-error theorem is developed and verified for the ecological vector field, Corollary S5.6 is stated and used only as a linearized, fixed-input impossibility result.

S5.6. Mandatory modeling restriction

Theorem S5.5 and Corollary S5.6 force the paper to impose at least one of the following restrictions to make discrimination well-posed:

1. A fixed maximum latent dimension m (the route taken in this paper);
2. A lower bound or spacing constraint on latent relaxation rates λ_j;
3. A complexity penalty or Bayesian prior that shrinks large-m models;
4. A specified experimental bandwidth and horizon that limits the information accessible to the model-selection procedure;
5. Auxiliary environmental measurements that constrain the latent mechanism from outside the input–output channel.

Without such a restriction, "fractional versus latent" is *not* a well-posed finite-data contest: with enough latent states, the latent model can mimic any finite-data signature of the fractional kernel to within the noise floor. Our choice is restriction (1): the latent model class $\mathcal{H}_m$ is parameterized by dimension m and a complexity budget, making the experiment design finite-dimensional and optimizable.

This restriction is not arbitrary, but neither is it a biological law. The latent-dimension cap $m_{\max}$ used in the experiments ($m \leq 5$) is a *declared experimental-model complexity budget*: it renders the discrimination contest finite-dimensional and optimizable, and it is motivated only by the practical observation that intermediate-complexity ecological descriptions tend to involve a small number of unmeasured slow variables. No theorem or dataset establishes $m \leq 5$ as a universal ecological bound. The cap is therefore treated as an experimental knob: sensitivity of every reported accuracy and design ranking to several values of $m_{\max}$ must be (and in the main-paper benchmark is) reported, and conclusions are stated relative to the declared budget.

S5.7. Frozen latent hierarchy and interval-enclosed approximation error

We freeze a single latent hierarchy for the finite-budget discrimination problem. Let T be the experiment horizon and $k_\alpha(t) = t^{\alpha-1}/\Gamma(\alpha)$ the fractional kernel. For each budget m define

$$\mathcal{K}_m = \left\{ k_m(t) = \sum_{j=1}^{m} c_j e^{-\lambda_j t} \,:\, c_j \geq 0,\ \lambda_j \in [\ell_m, L_m],\ \|k_m\|_{L^1(0,T)} \leq M_T \right\},$$

with zero-weight padding and genuinely nested running-envelope windows

$$\ell_m = \min_{j \leq m} \ell_j^*, \qquad L_m = \max_{j \leq m} L_j^*,$$

so $\mathcal{K}_m \subseteq \mathcal{K}_{m+1}$. For the atlas budgets $m = 4, 8, 16, 32$ these windows are approximately $[0.0206, 1.1956]$, $[0.0147, 1.5537]$, $[0.00742, 1.5537]$, and $[0.00742, 2.9913]$, respectively. The declared mass bound is $M_T = \|k_\alpha\|_{L^1(0,T)} + 1$. Every constructive competitor used numerically is checked against this mass bound; when a raw left-node mixture exceeds it, its coefficients are rescaled downward before its interval error enclosure is used. Theorem S5.3 separately guarantees arbitrarily accurate finite mixtures when the rate window is allowed to expand; no asymptotic claim is inferred solely from the finite numerical table below.

The kernel hierarchy is the adversarial object in the theorem that follows. Its connection to the full ecological state-space realization is deliberately kept distinct: the theorem is a linear convolution/observation statement. A window-uniform lift to the full predator–prey realization, including common safety over every $k_m \in \mathcal{K}_m$, is reserved for the atlas stage.

Definition S5.8 (Interval-enclosed constructive error)**.** For every stored explicit mixture $\widetilde{k}_m \in \mathcal{K}_m$, let U_m^{IA} be the interval-arithmetic upper enclosure produced by the released interval-arithmetic checker. The singular head $(0, \delta)$ is bounded by the exact component integrals

$$\int_0^\delta |k_\alpha - \widetilde{k}_m| \, dt \leq \int_0^\delta k_\alpha(t)\, dt + \int_0^\delta \widetilde{k}_m(t)\, dt,$$

and on every body subinterval $I_r \subset [\delta, T]$ interval arithmetic encloses $k_\alpha(t) - \widetilde{k}_m(t)$ for all $t \in I_r$, giving

$$\int_{I_r} |k_\alpha - \widetilde{k}_m| \, dt \leq |I_r| \, \sup |[k_\alpha - \widetilde{k}_m](I_r)|.$$

Because the classes are nested, define the best available constructive enclosure at budget m by

$$\widehat{E}_m^{\mathrm{IA}} := \min_{j \leq m} U_j^{\mathrm{IA}}.$$

Then $E_m := \inf_{k \in \mathcal{K}_m} \|k_\alpha - k\|_{L^1(0,T)} \leq \widehat{E}_m^{\mathrm{IA}}$.

For $\alpha = 0.85$, $T = 12$, and $\delta = 10^{-4}$, the independently recomputed interval upper enclosures are

m	1	2	3	4	6	8	12	16	24	32
$\widehat{E}_m^{\mathrm{IA}}$	0.925	0.569	0.541	0.352	0.248	0.248	0.236	0.200	0.200	0.134

The original `scipy.integrate.quad` values were high-accuracy numerical estimates, but their returned QUADPACK error estimates are not proof certificates; they are therefore not used as theorem constants. The interval checker raises the corresponding finite-m bounds by only about 0.15%–1.06% while providing an enclosure by construction.

S5.8. Minimum-error bound for safe memory discrimination

Theorem S5.9 (Testing bound under bounded safe excitation)**.** *Let $S : L^2(0,T) \to \mathbb{R}^q$ be a bounded linear observation operator and let the two simple observation laws be Gaussian with common positive-definite covariance R and means $S(k * u)$. Let $\mathcal{U}_m$ be any admissible common-safe input class satisfying $\|u\|_2 \leq B_{\mathrm{eff}}$ for all $u \in \mathcal{U}_m$, and define*

$$C_{\mathrm{obs}} := \frac{1}{2} \|R^{-1/2} S\|^2_{L^2 \to \mathbb{R}^q}, \qquad U_m := C_{\mathrm{obs}} \, (\widehat{E}_m^{\mathrm{IA}})^2 B_{\mathrm{eff}}^2.$$

For equal prior probabilities, define the robust binary minimax error

$$P_e^*(m) := \inf_{u\in\mathcal{U}_m} \inf_{\phi} \sup_{k\in\mathcal{K}_m} \frac{1}{2}\left[P^u_{C_\alpha}(\phi=1) + P^u_k(\phi=0)\right].$$

Then

$$P_e^*(m) \ge \Psi(U_m),$$

where valid choices include

$$\Psi_{\mathrm{Pin}}(x) = \max\left\{0, \frac{1-\sqrt{x/2}}{2}\right\}, \qquad \Psi_{\mathrm{BH}}(x) = \frac{1-\sqrt{1-e^{-x}}}{2},$$

and, for the equal-covariance Gaussian two-point reduction,

$$\Psi_{\mathrm{G}}(x) = \Phi(-\sqrt{x/2}).$$

Proof idea. Young's inequality bounds the mean difference of the two-point Gaussian subproblem by $\widehat{E}_m^{\mathrm{IA}} B_{\mathrm{eff}}$; the Gaussian KL bound, Pinsker / Bretagnolle–Huber, and uniformity of the KL ceiling over the input class complete the argument. Complete proof in the Supplementary Material (Section S2). □

The theorem is an observation-model theorem, not an identification of continuous white-noise observation with the point-sampled benchmark. Under the normalized specialization $\|S\| \le 1$ and $R = \sigma^2 I$, $C_{\mathrm{obs}} \le 1/(2\sigma^2)$. With the universal peak cap $B_{\mathrm{eff}} = \sqrt{T}\,u_{\max} = 0.346$ and $\sigma = 0.10$, the interval constants give, for example, $U_{16} \le 0.240$ and the Pinsker bound $P_e^*(16) \ge 0.327$; at $m = 32$, $U_{32} \le 0.108$ and $P_e^*(32) \ge 0.384$. These numbers demonstrate non-vacuity for the stated normalized Gaussian experiment. They are not benchmark error guarantees: the released benchmark uses discrete point samples and a fixed rival set, and remains a separate empirical validation layer.

The universal statement uses only the actuator peak cap and therefore applies to every common-safe subset. The Strong-Allee-specific tightening of the next subsection supplies the missing theorem-valid, hierarchy-uniform outer bound.

S5.9. Hierarchy-uniform Strong-Allee budget

In the ecological realization of the frozen hierarchy, each kernel $k(t) = \sum_j c_j e^{-\lambda_j t} \in \mathcal{K}_m$ acts through the ODE backbone $J(A)$ on the prey channel,

$$\dot{q}_j = -\lambda_j q_j + u, \qquad \dot{\xi} = J\xi + e_1 \sum_j c_j q_j,$$

so the prey deviation is $x_k = h_J * (k * u)$ with $h_J(t) = e_1^\top e^{Jt} e_1$. The state response is linear in the positive mode measure $\sum_j c_j \delta_{\lambda_j}$. The downward-gain functional $d(k) = \max_t [-x_k(t)]_+/\|u_0\|_2$ is sublinear rather than linear; nevertheless, positivity of the mode weights makes its supremum over the whole class reducible to a single-mode extremizer.

Theorem S5.10 (Hierarchy-uniform Strong-Allee budget)**.** *For a fixed protocol shape u_0 define*

$$F(\lambda) = \max_{t\le T} \frac{[-(h_J * (e^{-\lambda\cdot} u_0))(t)]_+}{\|u_0\|_2}, \qquad w(\lambda) = \frac{1-e^{-\lambda T}}{\lambda}.$$

Then the maximal downward gain of the entire class $\mathcal{K}_m$ is

$$d_{\mathrm{rob}}(\mathcal{K}_m, u_0) = M_T \max_{\lambda\in[\ell_m, L_m]} \frac{F(\lambda)}{w(\lambda)},$$

and the maximum is attained by the single-mode kernel $k^*(t) = \frac{M_T}{w(\lambda^*)} e^{-\lambda^* t}$ with $\lambda^* \in \arg\max F/w$. Consequently every common-safe ray $u = cu_0$ satisfies the necessary outer bound

$$\|u\|_2 \;\le\; B_{\mathrm{Allee}}(A, m) := \frac{\rho_\eta}{d_{\mathrm{rob}}(\mathcal{K}_m, u_0)}, \qquad \rho_\eta = x^* - (A+\eta),$$

which depends on A and m. This is a bound for the declared ray $u = cu_0$; it may be inserted into Theorem S5.9 only when that theorem's admissible input class is restricted to the same protocol ray. It is not an outer-energy bound for arbitrary common-safe waveforms.

Proof idea. Linearity of the state response in the positive mode measure bounds the downward gain by a mass-weighted maximum of F/w, attained by the single-mode kernel of full mass at the maximizing rate. Complete proof in the Supplementary Material (Section S2). □

Corollary S5.11 (Protocol-restricted testing bound)**.** *Let $\mathcal{U}_m(u_0)$ be the common-safe positive ray $\{cu_0 : c \geq 0\}$ intersected with the actuator peak cap. Then Theorem S5.9 applies with*

$$B_{\mathrm{eff}}(A, m; u_0) = \min\left\{B_{\mathrm{Allee}}(A, m; u_0),\ u_{\max}\frac{\|u_0\|_2}{\|u_0\|_\infty}\right\}.$$

No claim is made here for the minimax error over unrestricted safe waveform classes.

$F(\lambda)$ depends on the backbone $J(A)$ and the protocol, so d_{rob} has the exact variational form above. The displayed numerical constants are obtained from high-resolution time/rate grids and cross-resolution checks; unlike the interval enclosure for $\widehat{E}_m^{\mathrm{IA}}$, this maximization is not an outward-rounded interval certificate. For the pulse protocol, $\alpha = 0.85$, $T = 12$, the corrected nested-window grid gives lower estimates $d_{\mathrm{rob}} \approx 7.31$ at $(A, m) = (0.25, 4)$ and 22.20 at $(0.34, 32)$; the machine-readable values are reported in the released validation artifacts. For the pulse ray, the active budget $B_{\mathrm{eff}} = \min(B_{\mathrm{Allee}}, B_{\mathrm{cap}}, \sqrt{T}\,u_{\max})$ switches between the sampled values $A = 0.15$ and $A = 0.20$ for every displayed m; this is a numerical bracket, not an interval-certified root enclosure; for $A \geq 0.38$ the backbone response weakens near the stability edge and B_{Allee} increases again, so no monotonicity in A is asserted. Inserted through Corollary S5.11, this couples fractional approximation error, latent complexity, and Strong-Allee common safety over the same latent hierarchy for the declared pulse protocol. The fixed-rival crossover $A \approx 0.32$ of Section S7.7 remains a locked-set diagnostic; the hierarchy-uniform pulse-ray crossover is bracketed by the sampled values 0.15 and 0.20.

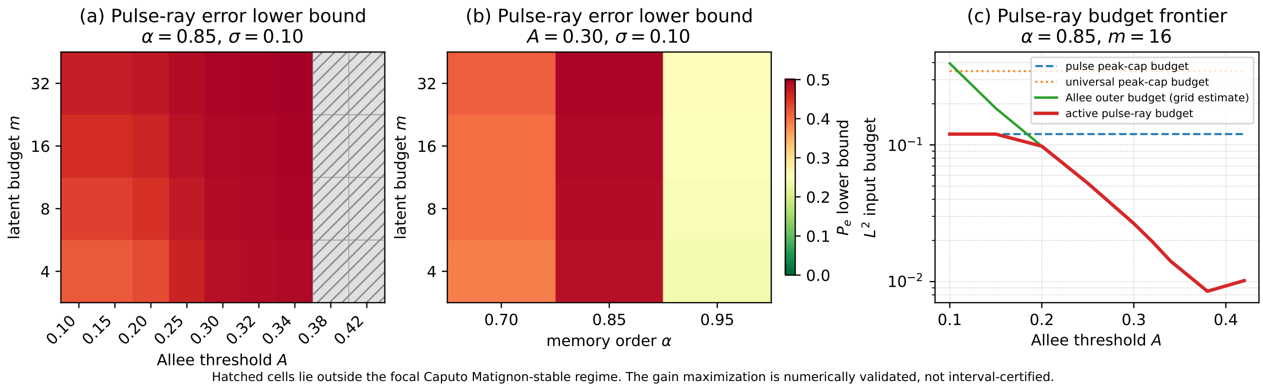

Figure S6. Protocol-restricted safe memory-discrimination atlas. Panels (a)–(b) show the Gaussian two-point lower bound for the declared positive pulse ray, using mass-valid interval enclosures of the constructive kernel error and the hierarchy-uniform Strong-Allee outer budget. Hatched cells in panel (a) lie outside the focal Caputo Matignon-stable regime and are not included in the headline stable-regime interpretation. Panel (c) shows the pulse-ray budget frontier; the Strong-Allee extremal identity of Theorem S5.10 is exact, while the displayed maximization constants and crossover are high-accuracy numerical estimates rather than interval-certified quantities. An inconclusive lower bound is never labeled discriminable.

S5.10. A certified ecological prey-response theorem via a backbone-consistent finite-state realization

The preceding results bound discrimination at the abstract memory-kernel level. A naïve ecological lift based on the false factorization $(s^\alpha I - J)^{-1} = (sI - J)^{-1}s^{-\alpha}$ is obstructed by a non-vanishing backbone mismatch. We therefore work directly with the true prey-to-prey transfer of the Caputo linearization,

$$G_\alpha(s) = e_1^\top (s^\alpha I - J(A))^{-1} e_1, \qquad g_\alpha(t) = e_1^\top t^{\alpha-1} E_{\alpha,\alpha}(J(A)t^\alpha) e_1.$$

For the complex-conjugate eigenvalues $\lambda, \bar{\lambda}$ of $J(A)$ and $J_{22} = 0$,

$$g_\alpha(t) = w_1 e_\alpha(t;\lambda) + \overline{w_1} e_\alpha(t;\bar{\lambda}), \qquad w_1 = \frac{\lambda}{\lambda - \bar{\lambda}},$$

where $e_\alpha(t;\lambda) = t^{\alpha-1}E_{\alpha,\alpha}(\lambda t^\alpha)$. Pole-aware contour representations and rational/finite-state approximations of Mittag–Leffler responses are established numerical tools [28, 31]; the contribution below is not the Hankel deformation itself, but its use to construct an interval-certified finite-state approximation of the ecological prey channel and to propagate that enclosure into a testing bound.

Lemma S5.12 (Pole/branch-cut decomposition)**.** *Let* $0 < \alpha < 1$ *and* $\lambda \in \mathbb{C}$ *satisfy* $\alpha\pi/2 < |\arg\lambda| < \alpha\pi$. *Then, for* $t > 0$,

$$e_\alpha(t;\lambda) = \frac{s_0}{\alpha\lambda}e^{s_0 t} + \int_0^\infty e^{-rt}\varphi_\lambda(r)\,dr, \qquad s_0 = \lambda^{1/\alpha},$$

with the principal branch and

$$\varphi_\lambda(r) = \frac{1}{\pi}\,\frac{r^\alpha \sin(\alpha\pi)}{r^{2\alpha} - 2\lambda r^\alpha \cos(\alpha\pi) + \lambda^2}.$$

Within this sector, the Matignon inequality $|\arg\lambda| > \alpha\pi/2$ *is equivalent to* $\operatorname{Re} s_0 < 0$*; the additional inequality* $|\arg\lambda| < \alpha\pi$ *guarantees that this principal-sheet pole exists.*

Proof idea. Deform the Bromwich contour onto the Hankel contour around the negative real axis; the sector hypothesis yields exactly one principal-sheet pole with the stated residue, and the two rays combine into the branch integral with a zero-free denominator. Complete proof in the Supplementary Material (Section S2). □

Summing the conjugate pair gives a real decomposition $g_\alpha = p + b$,

$$p(t) = 2\operatorname{Re}[Pe^{s_0 t}], \qquad b(t) = \int_0^\infty e^{-rt}\Phi(r)\,dr, \qquad \Phi(r) = 2\operatorname{Re}[w_1\varphi_\lambda(r)].$$

The pole term is exactly realizable by a stable real two-state integer-order block. The remainder is a generally signed Laplace-mixture density. Thus the earlier factorized-route obstruction is bypassed without asserting that the branch remainder is itself a positive or completely monotone kernel.

Finite-state response class. For each budget $m \geq 3$ and nondecreasing rate cutoff R_m, let $\mathcal{G}_m$ denote strictly proper, stable, finite-dimensional real LTI prey-response surrogates of state dimension at most m, with all poles in $\{\operatorname{Re} s < 0,\ |s| \leq \max(R_m, |s_0|)\}$ and impulse response satisfying $\|g\|_{L^1(0,T)} \leq M_{\text{state}}(A)$, where M_{state} is a certified upper enclosure of $\|g_\alpha\|_{L^1(0,T)}$ plus one. This is a response-level latent-surrogate class; it is not claimed that every member preserves the original predator–prey Jacobian skeleton. The constructive member consists of the exact two-state pole block plus $m - 2$ stable first-order modes approximating the branch integral.

Theorem S5.13 (Certified ecological prey-response approximation)**.** *At every displayed Matignon-stable cell and for* $m \in \{4, 8, 16, 32\}$ *there is an explicit* $g_m \in \mathcal{G}_m$ *such that*

$$\|g_\alpha - g_m\|_{L^1(0,T)} \leq \widehat{E}_m^{\text{state}}(A).$$

The enclosure is obtained by adaptive interval subdivision of the explicit branch density on compact rate intervals and the tail bound

$$\int_0^T \left|\int_{R_m}^\infty e^{-rt}\Phi(r)\,dr\right| dt \leq \frac{2|w_1|\sin(\alpha\pi)}{\pi\kappa\alpha}R_m^{-\alpha}, \quad \kappa = 1 - 2|\lambda||\cos(\alpha\pi)|R_m^{-\alpha} - |\lambda|^2 R_m^{-2\alpha} > 0.$$

For each finite body interval $[a, b]$, *the quadrature error is bounded by the minimum of a monotonicity bound and a second-order Taylor-moment bound, with all compact-interval integrals enclosed by interval arithmetic. No floating-point quadrature error estimate is used as a certificate.*

Proof. Use the exact pole/branch decomposition and realize the pole pair exactly. On each rate interval, split the branch error into $(e^{-rt} - e^{-\hat{r}t})\Phi(r)$ plus the interval-enclosed weight error at $\hat{r}$. Monotonicity of $w(r) = \int_0^T e^{-rt}dt$ gives the first local bound; Taylor expansion in r with an exact remainder gives the second. The tail follows from $\int_0^T e^{-rt}dt \leq 1/r$ and the reverse-triangle lower bound on the branch denominator. For convergence, first fix a finite rate cutoff and refine the partition, then let the cutoff grow; a diagonal choice gives $\widehat{E}_m^{\text{state}} \to 0$. Once the certified error is below one, the declared mass cap also follows directly from $\|g_m\|_1 \leq \|g_\alpha\|_1 + \|g_m - g_\alpha\|_1$; the released finite-budget constructions additionally pass the explicit mass check. □

For $\alpha = 0.85$ and $T = 12$, the certified focal enclosures are

m	4	8	16	32
$\widehat{E}_m^{\text{state}}$, $A = 0.25$	0.5335	0.1254	0.0274	0.0070
$\widehat{E}_m^{\text{state}}$, $A = 0.30$	0.4706	0.0899	0.0204	0.0067

An independent floating-point reconstruction using the Mittag–Leffler series on a transformed time grid agrees with all eight focal enclosures; the observed numerical L^1 distances are approximately 19%–38% of the corresponding interval upper bounds.

Corollary S5.14 (Prey-response testing bound)**.** *Let $S : L^2(0,T) \to \mathbb{R}^q$ be a bounded linear observation operator on the prey-deviation channel, let the Gaussian covariance be R, and set $C_{\mathrm{obs}} = \frac{1}{2}\|R^{-1/2}S\|^2$. For any input class with $\|u\|_2 \le B_{\mathrm{eff}}$, the robust minimax error for testing the fractional prey response against $\mathcal{G}_m$ satisfies*

$$P_e^{*,\mathrm{state}}(m) \ge \Psi\Big(C_{\mathrm{obs}}(\widehat{E}_m^{\mathrm{state}})^2 B_{\mathrm{eff}}^2\Big),$$

with the same testing inequalities used in Theorem S5.9.

Proof. Young's inequality gives $\|(g_\alpha - g_m) * u\|_2 \le \widehat{E}_m^{\mathrm{state}} B_{\mathrm{eff}}$. Insert this explicit competitor into the equal-covariance Gaussian two-point subproblem and then lower-bound the composite minimax risk by that subproblem. □

For the declared rectangular pulse and peak cap $\|u\|_\infty \le 0.10$, $B_{\mathrm{eff}} \le 0.10\sqrt{1.44} = 0.120$. At $\sigma = 0.10$, the resulting Pinsker lower bounds at $(A, m) = (0.25; 4, 8, 16, 32)$ are approximately $0.340, 0.462, 0.492, 0.498$, and at $A = 0.30$ they are $0.359, 0.473, 0.494, 0.498$. These values use only the physical pulse peak cap and therefore do not depend on a hierarchy-wide ecological safety extremizer.

Remark S5.15 (Why the broad mass-only Strong-Allee sandwich is not used)*.* A mathematically valid outer ray bound can be obtained over a broader mass-only response class by inserting the witness $g(t) = -ae^{-rt}$. That class, however, imposes no independent residue/gain cap and does not require members to retain the exact pole block of the approximation construction. Consequently the witness can use very large residues at large rate while preserving its finite-horizon L^1 mass. We therefore do not use that broader class to claim a hierarchy-wide ecological-safety closure. The prey-response testing result above uses the physical actuator/shape cap; the Strong-Allee hierarchy result of Theorem S5.10 remains the paper's ecological-safety result at kernel level.

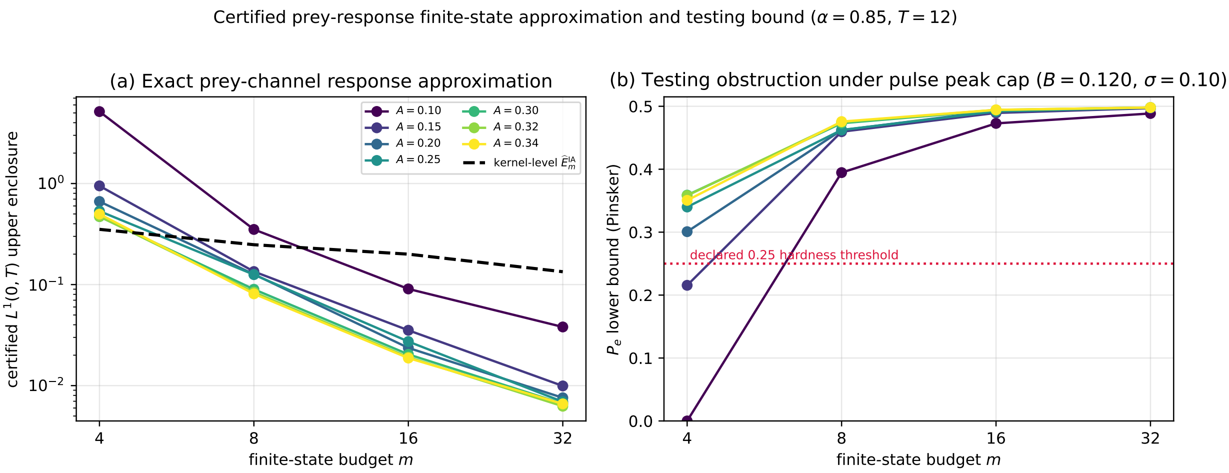

Figure S7. Certified finite-state approximation of the exact fractional prey response. Panel (a): outward-rounded $L^1(0,T)$ enclosures $\widehat{E}_m^{\mathrm{state}}$ across the Matignon-stable $\alpha = 0.85$ cells, compared with the earlier kernel-level enclosures. Panel (b): the corresponding Pinsker testing lower bound for the declared pulse under the physical peak cap $\|u\|_\infty \le 0.10$ ($B \le 0.120$) and $\sigma = 0.10$. The figure intentionally does not use the broad mass-only Strong-Allee witness; see Remark S5.15.

The next section formalizes optimal experiment design using the finite-horizon bounds developed here.

S6. Optimal Input and Observation Design

The structural separation of Section S4 and the finite-horizon impossibility barrier of Section S5 together define the playing field. The fractional model is generically distinguishable from every finite latent or delay model (Theorem S4.5), but on any finite horizon a sufficiently rich latent model can mimic the fractional output below the noise floor (Corollary S5.6). The resolution adopted in Section S5 is to bound the latent dimension by a complexity budget m, which makes the discrimination contest a finite-dimensional optimization. This section solves that optimization [1, 2, 11, 13, 20, 26, 35]: given a model pair, an energy budget, and a measurement noise covariance, we construct

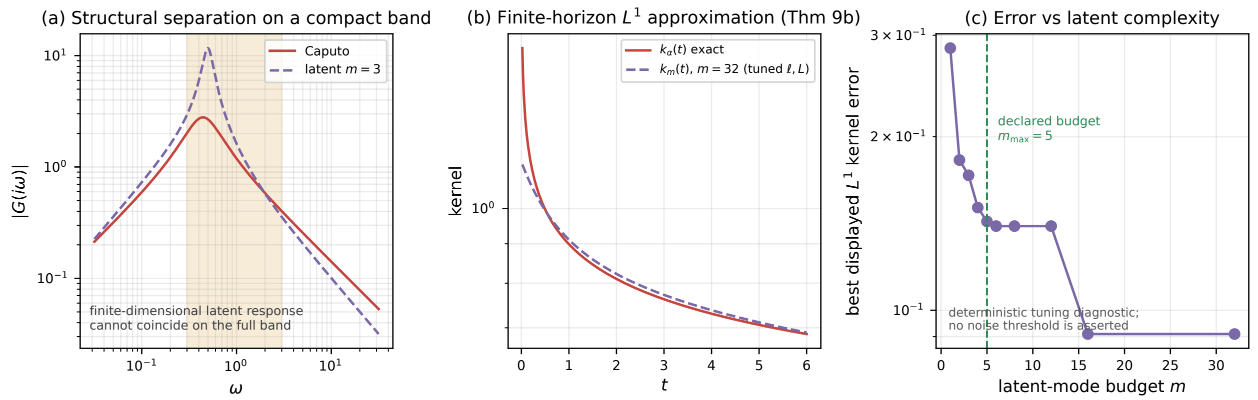

Figure S8. Exact separation versus finite-horizon approximation, in one picture. Panel (a): on a compact band the Caputo and finite-latent transfer magnitudes agree only at finitely many frequencies (Theorems S4.1–S4.5); panel (b): a tuned display of the constructive exponential-sum family of Theorem S5.3 tracks the singular kernel k_α in $L^1(0,T)$ (here $m = 32$); panel (c): the deterministic best-displayed L^1 approximation error decreases as the latent-mode budget grows. The green line marks the declared modeling budget $m_{\max} = 5$; no statistical noise threshold is asserted in this panel. The theoretical fixed-noise-floor consequence is stated separately in Corollary S5.6.

the excitation and the sampling schedule that maximize the pairwise Kullback–Leibler divergence [7, 10, 18, 21, 37] under explicit constraints. The construction is exact for the linear-Gaussian regime and supplies the benchmark against which any constrained implementation (box, rate, or cumulative-impact) is measured.

The section is organized around the operator ΔH that measures the inter-model output gap. Input design asks how to drive $\Delta H u$ as large as possible per unit energy; observation design asks where in time and channel to measure so that the realized gap is best converted into information. The two are dual and we develop them in sequence.

S6.1. Pairwise linear-Gaussian formulation

We work in the linearized regime of Section S3. Fix two candidate models mapping an input u to a finite observation vector

$$Y_i = H_i u + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, R), \qquad i \in \{1, 2\}, \tag{S16}$$

where H_i is the (input-to-observation) operator of model i — a finite-horizon discretization of the convolution with the channel G_i of Section S3 — and R is the symmetric positive definite measurement noise covariance. Define the inter-model difference operator

$$\Delta H := H_1 - H_2. \tag{S17}$$

Under the Gaussian assumption, the pairwise Kullback–Leibler divergence between the two output distributions induced by a common input u admits the closed form

$$D_{12}(u) = \tfrac{1}{2}\|R^{-1/2}\Delta H u\|^2, \tag{S18}$$

independent of the mean offset and fully determined by how much the input, filtered through ΔH, escapes the noise cone defined by R. The design problems that follow are all instances of maximizing (S18) under constraints on u, and subsequently on the observation locations.

S6.2. Theorem T11 – energy-constrained optimal input

The first design question is: under a total energy budget $\|u\|^2 \le E$, how large can the pairwise divergence be made, and which input achieves it?

Theorem S6.1 (Energy-optimal pairwise input)**.** *Consider the pairwise model* (S16) *with ΔH as in* (S17) *and $R > 0$. Define the positive semidefinite information operator*

$$Q := \Delta H^* R^{-1} \Delta H. \tag{S19}$$

Then the maximal pairwise divergence under the energy constraint $\|u\|^2 \le E$ is

$$\boxed{D_{12}^{\max} = \frac{E}{2}\lambda_{\max}(Q),} \tag{S20}$$

and an optimal input is

$$\boxed{u^* = \sqrt{E}\, v_{\max},} \tag{S21}$$

where $v_{\max}$ is a unit principal eigenvector (or eigenfunction, in the input-function setting) of Q.

Proof idea. The divergence is the Rayleigh quotient of the information operator Q under the energy constraint; the maximum is attained on the principal eigenspace. Complete proof in the Supplementary Material (Section S3). □

Remark S6.2. Theorem S6.1 solves the pairwise linear-Gaussian design exactly. The optimal input is the principal eigenfunction of the noise-weighted inter-model operator, not a generic excitation: a pulse, chirp, PRBS, or multisine is optimal only to the extent that it approximates $v_{\max}$ under additional amplitude, sign, duration, or implementation constraints. The principal eigenfunction concentrates energy precisely where the two transfer functions disagree most strongly relative to the noise floor — the frequency-domain reading of the same statement is developed in Section S6.3 below.

S6.3. Frequency-domain reading and the single-frequency optimum

For scalar stable LTI models with transfer difference $\Delta G(i\omega) = G_1(i\omega) - G_2(i\omega)$, white output noise of variance σ^2, and an input spectral energy measure ξ on a measurable frequency set Ω, the divergence becomes

$$D_{12}(\xi) = \frac{1}{2\sigma^2}\int_\Omega |\Delta G(i\omega)|^2\, d\xi(\omega), \qquad \xi(\Omega) \le E, \tag{S22}$$

which is the diagonal form of Theorem S6.1 in the sinusoidal basis. The information operator reduces to the scalar frequency score $|\Delta G(i\omega)|^2/(2\sigma^2)$.

Corollary S6.3 (Optimal single frequency)**.** *Under the setup of* (S22) *with fixed, distinct scalar models and a single-frequency constraint, the optimum is*

$$\omega^* \in \arg\max_{\omega\in\Omega} |\Delta G(i\omega)|^2,$$

and all available energy is assigned to ω^*. For a real-valued input, use the conjugate pair $\pm\omega^*$, i.e. a sinusoid at ω^*.

Remark S6.4. Corollary S6.3 concerns *fixed simple models*. It does *not* imply that a single sinusoid is optimal for parameter estimation or for robust discrimination among several composite (e.g. parameter-tunable latent) models: an adaptive latent model can match the fractional response at one frequency by tuning its internal parameters, exactly as anticipated in the discussion following Theorem S4.5. Robust discrimination requires broadband excitation, developed next.

For MIMO channels with frequency-dependent output covariance $R(\omega)$, the per-frequency score generalizes to

$$w_{12}(\omega) = \lambda_{\max}\Big(\Delta G(i\omega)^* R(\omega)^{-1} \Delta G(i\omega)\Big), \tag{S23}$$

and the optimal excitation at the selected frequency lies in the associated right singular direction. This is the frequency-domain avatar of the principal-eigenfunction prescription of Theorem S6.1.

S6.4. Theorem T12 – robust multisine design has finite support

Discriminating among P candidate model pairs simultaneously requires broadband excitation. Indexing the pairs by $p = 1, \ldots, P$ and writing the per-pair per-frequency score as

$$w_p(\omega) = \frac{1}{2\sigma^2}\left|G_{i_p}(i\omega) - G_{j_p}(i\omega)\right|^2, \tag{S24}$$

the robust spectral design is the linear program over nonnegative spectral measures

$$\max_{\xi,t}\ t \qquad \text{s.t.} \qquad \int_\Omega w_p(\omega)\, d\xi(\omega) \ge t, \quad p = 1, \ldots, P, \qquad \xi(\Omega) \le E, \quad \xi \ge 0. \tag{S25}$$

Problem (S25) optimizes the worst-case pairwise divergence (a maximin criterion) over an arbitrary continuous spectrum. The next theorem bridges the gap between this continuous ideal and any implementable experiment: an optimal measure *exists* and is supported on finitely many frequencies.

Theorem S6.5 (Finite multisine sufficiency)**.** *Assume $\Omega \subset \mathbb{R}$ is compact and each $w_p : \Omega \to \mathbb{R}_{\geq 0}$ is continuous. Then problem* (S25) *admits an optimal solution, and there exists an optimal spectral measure ξ^* supported on at most*

$$\boxed{|\operatorname{supp}\xi^*| \leq P+1} \tag{S26}$$

frequencies. Consequently, the robust broadband optimum is a multisine with no more than $P+1$ tones.

Proof idea. The robust spectral problem is a linear program over spectral measures whose feasible set is determined by P continuous score constraints on a compact band; Carathéodory's theorem bounds the support of an extreme-point solution by $P+1$ atoms. Complete proof in the Supplementary Material (Section S3). □

Remark S6.6. Theorem S6.5 gives the rigorous robust counterpart of the heuristic "use multiscale excitation": the rigorous object is a finite multisine whose frequencies and weights solve the linear program (S25). A logarithmic pulse train or chirp is an implementation surrogate, appropriate when fine-grained sinusoidal control is unavailable but a broadband spectral footprint is still desired. The bound $P+1$ is a worst-case Carathéodory guarantee; in many applications the optimum uses substantially fewer tones.

The connection back to Section S4 is direct. Theorem S4.5 guarantees that almost every frequency separates any *fixed* pair, but a parameter-tunable latent model can close the gap at any single frequency. Theorem S6.5 addresses the fixed-pair version of this tension rigorously: by spreading energy across at most $P+1$ carefully chosen tones, the design guarantees a uniform lower bound t^* on the divergence of *every fixed* pair simultaneously. Two scope limits must be kept in view. First, P counts the fixed score functions entering (S25); for a composite rival class indexed by continuous parameters the maximin problem is semi-infinite, and the $P+1$ support bound applies only after a justified finite reduction or discretization of the parameter set. Second, a finite multisine does not, by itself, prevent a parameter-tunable latent family from fitting all selected frequencies jointly; what it removes is the single-frequency escape route, not adaptivity in general. Together with the complexity bound m of Section S5 and the exclusions of Remark S5.7, this converts the structural separation into a finite, computable discrimination barrier for the model classes of this paper.

S6.5. Theorem T13 – optimal pairwise sampling schedule

Input design fixes what is excited; observation design fixes what is measured. Fix the input u selected above and consider the placement of n measurements on a finite candidate grid of time–channel pairs (t, o). Assume independent Gaussian measurement noise of variance $\sigma_o^2(t)$ at each candidate, and define the *normalized separation score*

$$s_{12}(t,o) := \frac{\big(\mu_1(t,o) - \mu_2(t,o)\big)^2}{2\,\sigma_o^2(t)}, \tag{S27}$$

where $\mu_i(t,o)$ is the noise-free mean response of model i at time t on channel o. For a selected schedule S of independent measurements, Gaussanity makes the divergence additive:

$$D_{12}(S) = \sum_{(t,o)\in S} s_{12}(t,o). \tag{S28}$$

Theorem S6.7 (Optimal pairwise sampling schedule)**.** *Suppose exactly n distinct measurements may be selected from a finite candidate set, the measurements are independent given the model, and there are no coupling constraints. Then the KL-optimal schedule consists of the n candidates with the largest separation scores $s_{12}(t,o)$.*

Proof. The objective (S28) is additive over S. If a schedule S omits a candidate $c^\dagger$ whose score exceeds that of some selected candidate $c \in S$, then replacing c by $c^\dagger$ strictly increases $D_{12}(S)$, contradicting optimality. Hence every optimal schedule is a top-n set. Uniqueness holds up to ties at the threshold score. □

Remark S6.8. If repeated observations at the same candidate are permitted, with iid noise and no saturation, the degenerate optimum places all n observations at a global score maximum. In practice, spacing constraints, temporal correlation, or biological cost prevent this degeneracy; the spacing-constrained case is addressed in the Supplementary Material (Section S3).

S6.6. Practical layers: spacing, cost, channels, and constrained implementation

Three practical layers complete the design theory and are developed in full in the Supplementary Material (Section S3): (i) observation design under a minimum-spacing constraint (dynamic programming) and under per-sample costs and budgets (knapsack relaxation), together with channel selection from the strong-Allee linearization; (ii) the structural contrast between discrimination designs and parameter-estimation designs — the two objectives are driven by different operators and their optima generically differ; and (iii) constrained implementation of the optimal inputs under box, rate, and cumulative-impact constraints, where the unconstrained optimum of Theorem S6.1 remains the reference upper bound against which any feasible design is scored.

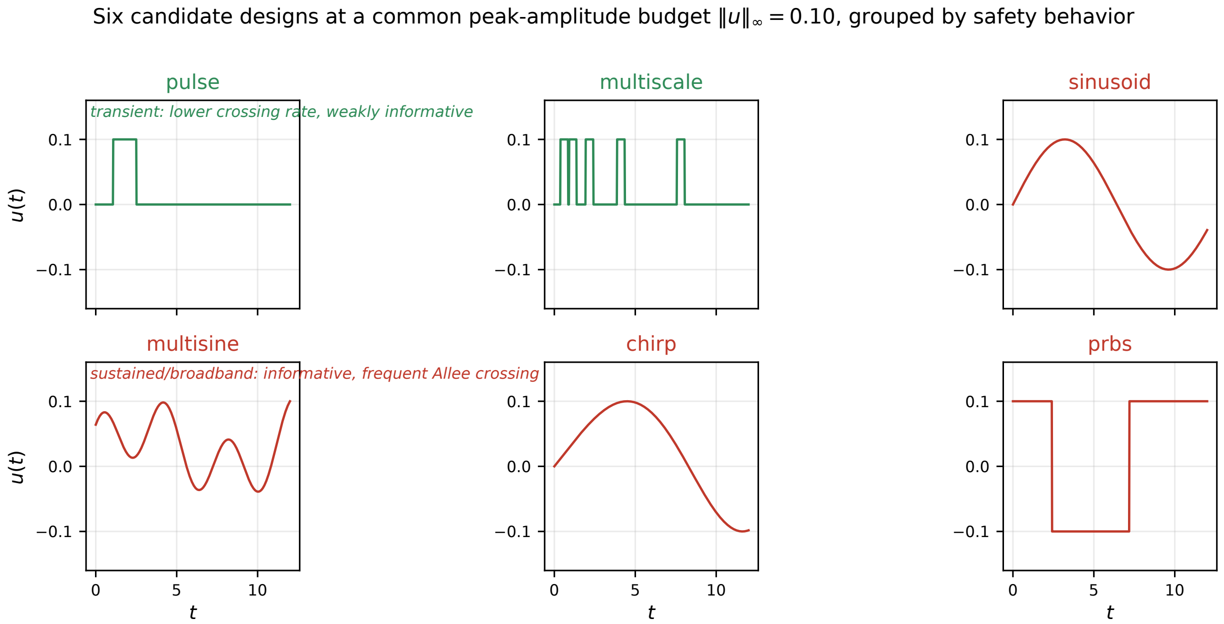

Figure S9. The six candidate input designs at the common peak-amplitude budget $\|u\|_\infty = 0.10$, grouped by their benchmark safety behavior (the main-paper benchmark): transient families (pulse, multiscale) have the lowest crossing rates in the benchmark but discriminate weakly; sustained/broadband families (sinusoid, multisine, chirp, prbs) are more informative yet cross the Allee threshold much more frequently.

S7. Safety-Constrained Experiment Design

The optimal input of Theorem S6.1 and the optimal schedule of Theorem S6.7 maximize discrimination information under energy and cardinality constraints; they do not, by themselves, guarantee that the experiment respects the ecological integrity of the system. Small-amplitude perturbations near the Allee extinction threshold can trigger irreversible collapse, and any actuator bound under that threshold violates the standing assumption of Section S1 that the state remains within the linear trust region around the coexistence equilibrium z^*. This section establishes explicit conditions under which an informative input is simultaneously safe [3, 9, 12, 19, 25, 36, 38]. Two results are central: a general existence theorem proving that a safe and informative perturbation always exists when the equilibrium has positive interior margin (Theorem S7.1), and a constructive fractional invariance theorem that translates the strong-Allee safety conditions into checkable linear inequalities on the input amplitude and its integral (Theorem S7.4). Together they close the loop between the structural discrimination theory of Sections S4— S6 and the biological mandate of Section S1.

S7.1. The safe set and the safety margin

Let $\mathcal{S} \subset \mathbb{R}^n$ be a closed safe region and let the coexistence equilibrium z^* of Section S1 lie in its interior with margin

$$\rho = \mathrm{dist}(z^*, \partial\mathcal{S}) > 0. \tag{S29}$$

For the strong-Allee predator–prey ensemble, a minimal safety condition is

$$x(t) \geq x_{\min} > A, \tag{S30}$$

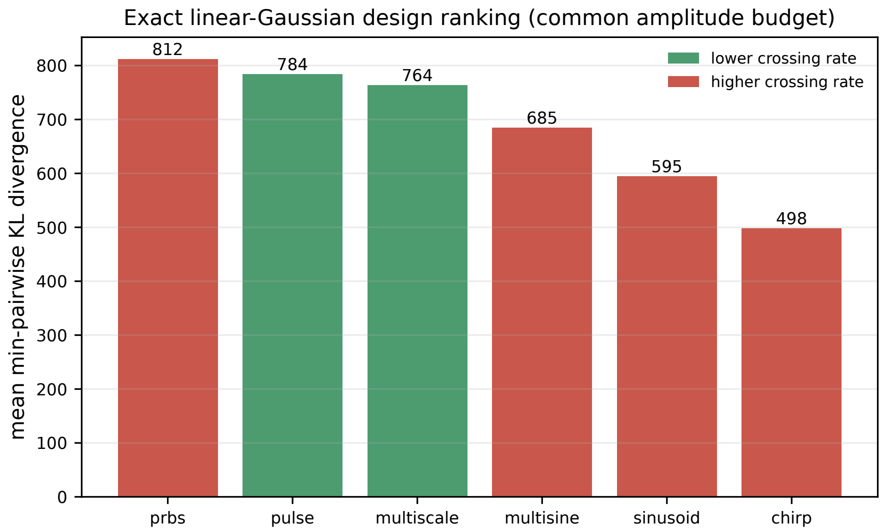

Figure S10. Exact linear-Gaussian design ranking by mean minimum pairwise KL divergence over the full factorial under the common peak-amplitude budget; state safety is not imposed in this ranking. Bars are colored by the nonlinear benchmark's safety verdict (the main-paper safety–informativeness figure): the linear optimum and the safe set disagree — the preview of the trade-off.

with additional upper-abundance and predator-vigor constraints as required by the basin of attraction described in Section S1. The Allee threshold A separates the extinction basin from the coexistence basin, and the validated paper [25] proves that trajectories crossing below $x = A$ enter an extinction funnel that no control budget can reverse. A safe protocol therefore reserves a certified margin above A and caps cumulative prey removal.

S7.2. Theorem T16 — existence of a safe and informative perturbation

Theorem S7.1 (Safe and informative perturbation exists)**.** *Consider two locally well-posed Caputo models of order $\alpha \in (0, 1]$ with the same safe equilibrium z^* and the same stable linearization J of Section S1. Let their input-to-observation operators over a finite horizon $[0, T]$ be H_1 and H_2, and assume:*

(i) the nonlinear solution maps $\Phi_u^{(1)}, \Phi_u^{(2)}$ are continuous with respect to the input amplitude on $[0, T]$;

(ii) an admissible direction v exists such that $(H_1 - H_2)v \neq 0$;

(iii) z^* has positive safety margin $\rho > 0$ as in (S29)*.*

Then there exists $\varepsilon_0 > 0$ such that every perturbation

$$u_\varepsilon = \varepsilon v, \qquad 0 < \varepsilon < \varepsilon_0, \tag{S31}$$

is safe for both models and yields strictly positive Kullback–Leibler divergence.

Proof idea. Interior margin plus continuity of the solution map in the input give a nonempty open set of safe perturbations; any nonzero member is informative for some pair by the separation results. Complete proof in the Supplementary Material (Section S4). □

Remark S7.2. Theorem S7.1 guarantees existence, but it does not guarantee that the resulting divergence exceeds an operational minimum $D_{\min}$ required for a prescribed confidence level. Given a minimal discrimination target $D_{\min}$, the feasibility condition becomes $\frac{1}{2}\varepsilon^2\langle v, Qv\rangle \geq D_{\min}$ where $Q = \Delta H^* R^{-1} \Delta H$ is the information operator of Theorem S6.1, and ε must remain below the safety ceiling ε_s. When the intersection is empty, the experiment is unresolvable under the specified safety budget; the designer must weaken one of the constraints or choose a more informative model pair.

S7.3. Linear finite-horizon safety bound

Theorem S7.1 is qualitative. The next result provides a quantitative, computable bound for the linearized Caputo system. Write the stable linear Caputo dynamics in the representation of Section S1:

$$\tau_0^{\alpha-1}\, {}^C D_t^\alpha \xi = J\xi + Bu, \qquad \xi(0) = 0, \tag{S32}$$

and define the impulse-response matrix

$$H_\alpha(t) = \tau_0^{1-\alpha} t^{\alpha-1} E_{\alpha,\alpha}\left(\tau_0^{1-\alpha} J t^\alpha\right) B, \tag{S33}$$

where $E_{\alpha,\alpha}(\cdot)$ is the two-parameter Mittag-Leffler function [23]. The state response to a bounded input is

$$\xi(t) = \int_0^t H_\alpha(t-s)\, u(s)\, ds. \tag{S34}$$

Define the finite-horizon gain

$$\Gamma_T := \sup_{0\le t\le T} \int_0^t \|H_\alpha(s)\|\, ds, \tag{S35}$$

where $\|\cdot\|$ denotes the operator norm induced by the safety metric (typically the Euclidean norm or a weighted norm aligned with the safe-set geometry).

Lemma S7.3 (Linear safety envelope)**.** *If the input satisfies the uniform bound*

$$\|u\|_{L^\infty(0,T)} \le \frac{\rho}{\Gamma_T}, \tag{S36}$$

then $\sup_{0\le t\le T} \|\xi(t)\| \le \rho$*, and the linearized solution remains within the safety margin of the equilibrium.*

Proof idea. Grönwall-type bounding of the Mittag-Leffler solution kernel on the finite horizon; complete proof in the Supplementary Material (Section S4). □

The bound (S36) is conservative (the true nonlinear margin is typically larger than the linear one), but is exact in the linear regime of Section S3 and provides a fast certificate: compute the impulse-response matrix $H_\alpha(t)$ for the Jacobian $J(A)$ and input channel B of Section S1, evaluate Γ_T numerically, and compare against the selected amplitude. When the fractional order α is close to 1, $H_\alpha(t)$ approaches the exponential $e^{Jt}B$ and Γ_T can be bounded by standard matrix-exponential estimates; for fractional $\alpha < 1$, the slower rational decay of the Mittag-Leffler kernel increases Γ_T and correspondingly tightens the amplitude ceiling.

S7.4. Theorem T17 — fractional inward-pointing rectangle

The linear bound of Lemma S7.3 is a local trust-region condition. For nonlinear safety, especially near the Allee threshold A where the prey dynamics flip sign, a stronger certificate is needed. The next result is a fractional generalization of the inward-pointing rectangle method [31]: if the vector field points strictly inward on every face of a rectangular box, the trajectory cannot reach the boundary and the interior of the box is positively invariant for any Caputo order $\alpha \in (0,1)$.

Theorem S7.4 (Fractional strict inward-pointing rectangle invariance)**.** *Consider the Caputo system*

$${}^C D_t^\alpha z = F(t,z), \qquad 0<\alpha<1, \tag{S37}$$

with F continuous in t, locally Lipschitz in z, and such that the initial-value problem has a unique continuous solution. Let

$$\mathcal{R} = \prod_{i=1}^n [\ell_i, u_i] \subset \mathbb{R}^n, \tag{S38}$$

and assume the initial condition lies in the interior, $z(0) \in \operatorname{int}\mathcal{R}$*. Suppose there exists a uniform margin* $\eta > 0$ *such that for all* $t \ge 0$*,*

$$F_i(t,z) \ge \eta \quad \textit{whenever} \quad z_i = \ell_i, \tag{S39}$$

$$F_i(t,z) \le -\eta \quad \textit{whenever} \quad z_i = u_i. \tag{S40}$$

Then the solution remains in the interior for all future times: $z(t) \in \operatorname{int}\mathcal{R}$ *for every* $t \ge 0$.

Proof idea. The Caputo endpoint identity ties the sign of the fractional derivative at a first boundary touching to the sign of the field at that point; strict inward-pointing on every face therefore contradicts a first exit. Complete proof in the Supplementary Material (Section S4). □

Remark S7.5 (Strictness and the boundary). The strict margin $\eta > 0$ and the interior initial condition are essential. Under the strict inequalities the first-contact contradiction is genuine; with only non-strict inequalities ($F_i \geq 0$ on lower faces, $F_i \leq 0$ on upper faces) the chain $0 \leq F_i = {}^C D_t^\alpha z_i \leq 0$ is *not* a contradiction, and boundary initial points require a separate viability argument. A non-strict invariance statement should be based on a cited fractional viability or invariance theorem with all hypotheses verified, together with an explicit treatment of initial points on the boundary; we do not develop that here and state only the strict version.

S7.5. Strong-Allee boundary inequalities

Applying Theorem S7.4 to the controlled strong-Allee system of Section S1 with additive controls u_x, u_y and a safety rectangle strictly above the Allee threshold yields four explicit, checkable face inequalities — lower and upper prey, lower and upper predator — bounding the admissible input amplitudes; near the Allee threshold the lower prey face is the binding constraint. The derivation of all four faces is given in the Supplementary Material (Section S4).

S7.6. Input envelope for a discriminative safest experiment

Combining Theorem S7.1 and Theorem S7.4 with the optimal designs of Theorem S6.1 and Theorem S6.5 yields a complete safety-certified design procedure for the linearized strong-Allee case:

1. Compute the unconstrained optimal input u^* from Theorem S6.1 (or, in the multisine case, from the solution of the linear program (S25) following Theorem S6.5).

2. Compute the linear safety gain Γ_T from (S35) using the impulse response (S33) and obtain the worst-case amplitude envelope (S36).

3. Project u^* onto the amplitude envelope: define the candidate safe input

$$u_{\text{safe}}(t) = \text{sat}_{\frac{\rho}{\Gamma_T}}(u^*(t)), \tag{S41}$$

   where $\text{sat}_M(v) = \min\{1, M/\|v\|\}\, v$ with the relevant norm. Pointwise saturation preserves feasibility but alters the spectral content of u^*; it is a feasible warm start and a certified lower bound on achievable information, *not* the optimizer of the box/rate/budget-constrained problem. The constrained optimum must be re-optimized (Supplementary Material, Section S3) and compared against the unconstrained upper bound of Theorem S6.1.

4. Verify that the projected input satisfies the four face inequalities of the Supplementary Material (Section S4). If a boundary fails, the design is infeasible at the chosen amplitude: reduce the amplitude (or, only with explicit ecological justification, enlarge the safety margin ρ); enlarging the box $\mathcal{R}$ itself weakens the safety guarantee and is not a mathematical remedy.

5. Execute the experiment with adaptive monitoring: if a single measurement falls below the safety floor, the protocol automatically terminates the trial and records the cause.

This five-step protocol crystallizes the claim of this section: the unconstrained optimal input of Section S6 remains the reference design; the safety layer caps its amplitude, verifies the face inequalities, and certifies feasibility. The amplitude-capped constrained optimum itself — maximizing discrimination information subject to the safety constraints rather than saturating pointwise — is the natural next problem and is left to future work.

The practical consequences of operating near the Allee threshold — which perturbation classes are intrinsically safer, how the safety margin trades against information, and what the threshold implies for actuator sizing — are discussed in the Supplementary Material (Section S4).

S7.7. Scope of safe-excitation bounds: a negative theorem and the corrected transient interface

The preceding certificates bound how strongly a safe experiment may excite the system. Two complementary facts complete the picture.

Theorem S7.6 (State safety alone does not bound input energy)**.** *Let $\mathcal{H}_x u = e_1^\top (H_\alpha * u)$ be the prey-channel map of the linearized Caputo system, and let the local safe class be $\mathcal{U}_{\text{safe}} = \{u \in L^2(0,T) : \inf_{t \leq T}[x^* + (\mathcal{H}_x u)(t)] \geq A + \eta\}$. Then $\sup_{u \in \mathcal{U}_{\text{safe}}} \|u\|_2 = +\infty$.*

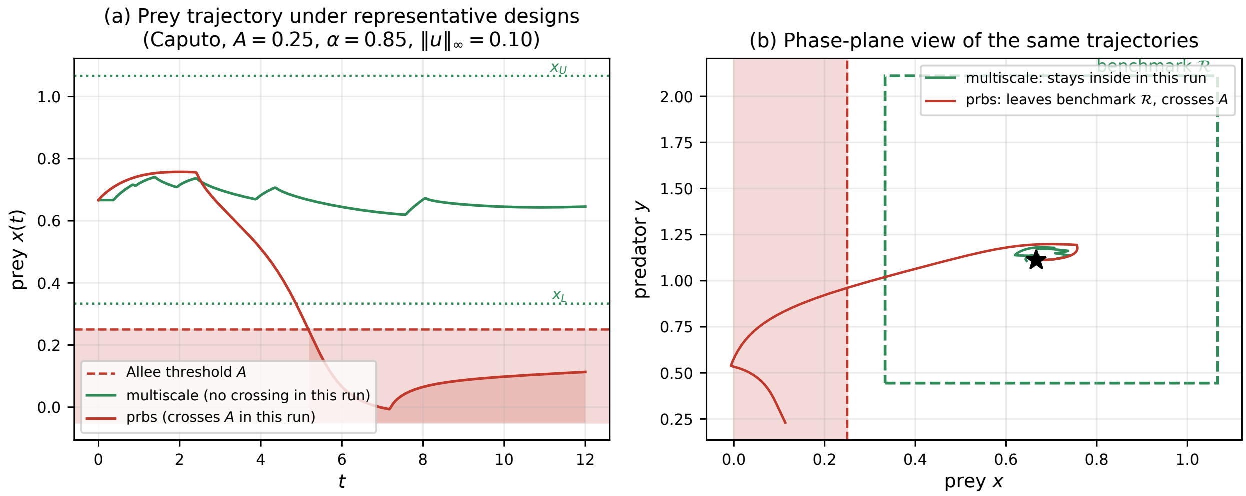

Figure S11. Safety geometry, computed with the released benchmark code for one representative stable-baseline cell ($A = 0.25$, Caputo $\alpha = 0.85$, $\|u\|_\infty = 0.10$). Panel (a): the multiscale trajectory does not cross the Allee threshold in this run (minimum margin +0.369), whereas prbs crosses it (minimum margin −0.257; shaded interval marks the violation). Panel (b): the same trajectories against the benchmark diagnostic rectangle $\mathcal{R}$. Remaining inside this rectangle in a simulation is a numerical diagnostic, not a proof of positive invariance; Theorem S7.4 provides such a certificate only when its strict face inequalities are verified. The figure visualizes the dynamical trade-off quantified over the full benchmark in the main-paper benchmark.

Proof. For $0 \le c \le 1$, scaling a safe input toward zero preserves the one-sided floor because $x^* > A+\eta$; no symmetry under negative scaling is asserted. Let $u_\omega(t) = \sqrt{2/T}\sin(\omega t)$, so $\|u_\omega\|_2 = 1 + O(1/(\omega T))$. Because the impulse-response coordinate $h_x = e_1^\top H_\alpha$ belongs to $L^1(0,T)$, the uniform Riemann–Lebesgue lemma for indefinite Fourier integrals gives

$$\sup_{t\le T}\left|\int_0^t h_x(r)e^{-i\omega r}\,dr\right| \xrightarrow{\omega\to\infty} 0,$$

hence $\|\mathcal{H}_x u_\omega\|_{L^\infty} \to 0$. For ω large enough the scaled input $(\rho_\eta/(2\delta_\omega))u_\omega$, with $\delta_\omega = \|\mathcal{H}_x u_\omega\|_\infty$ and $\rho_\eta = x^* - (A+\eta) > 0$, satisfies $x^* + (\mathcal{H}_x u)(t) \ge x^* - \rho_\eta/2 > A+\eta$ for every t, while its L^2 norm diverges as $\omega \to \infty$. □

Theorem S7.6 is the reason no Allee-dependent outer energy bound can hold for unrestricted inputs: high-frequency energy is invisible to a strictly proper memory channel. Any quantitative safe-discrimination bound therefore requires a declared input class — this is why Theorem S5.9 is stated relative to an explicit outer budget B_{eff} rather than to state safety alone. The observed decay $\|\mathcal{H}_x u_\omega\|_\infty \sim \omega^{-\alpha}$ for the probes used in the research rounds is a numerical diagnostic, not part of the theorem.

For the transient protocols of the main-paper benchmark the correct one-sided interface is as follows. Let $u = cu_0$ with $c \ge 0$ and $\|u_0\|_2 = 1$, and define the downward prey gain $d_M(u_0) = \max_t[-(\mathcal{H}_{x,M}u_0)(t)]_+$ for each rival memory law M. One-sided safety $x^* + c(\mathcal{H}_{x,M}u_0)(t) \ge A+\eta$ for all rivals implies the necessary bound

$$\|u\|_2 \le \frac{\rho_\eta}{d_{\text{rob}}}, \qquad d_{\text{rob}} = \sup_M d_M(u_0), \qquad \rho_\eta = x^* - (A+\eta),$$

which must be combined with the shape-specific consequence of the peak cap, $\|u\|_2 \le u_{\max}\|u_0\|_2/\|u_0\|_\infty$. Evaluated as numerical diagnostics on the benchmark realizations ($\alpha = 0.85$, $T = 12$, $\eta = 0.02$, $u_{\max} = 0.10$): the pulse shape has $d_{\text{rob}} = 1.18$ and shape cap 0.120 at $A = 0.25$, so the peak cap binds; the same holds at $A = 0.30$; at $A = 0.32$ the one-sided Allee bound (0.120) first ties the cap, and for $A \ge 0.34$ it binds strictly (0.085 at $A = 0.34$). The crossover $A \approx 0.32$ is a protocol-relative diagnostic of the locked benchmark rival set: proximity to the extinction threshold shrinks the admissible transient amplitude there. These d_{rob} values are not a supremum over the full hierarchy $\mathcal{K}_m$ and are not inserted into Theorem S5.9. The hierarchy-uniform version was subsequently obtained: Theorem S5.10 (Section S5.9) gives the exact supremum $d_{\text{rob}}(\mathcal{K}_m, u_0)$ over the entire class, attained by a single-mode kernel, and its budget $B_{\text{Allee}}(A, m)$ enters Theorem S5.9 after restricting the input class to that same fixed protocol ray. For the pulse protocol the hierarchy-uniform switch is numerically bracketed between $A = 0.15$ and $A = 0.20$, more conservative than the locked-set 0.32 diagnostic above, as it must be.

S8. Nonlinear Bayesian Extension

The structural discrimination theory of Sections S4–S6 and the safety-constrained design of Section S7 were all developed within the linear-Gaussian regime of Section S3. The optimality certificates of Theorem S6.1 (energy-optimal input) and Theorem S6.5 (finite multisine support) are exact in that regime, and the invariance certificate of Theorem S7.4 holds for the fully nonlinear Caputo dynamics. What they do not address is the question that closes the design loop: once data begin to arrive, which experiment should be run next, when the model is nonlinear, the noise is not Gaussian, and the parameters — including the Allee threshold A itself — are uncertain. This section bridges that gap [6, 15, 32, 33]. It states two results that pin the Bayesian layer analytically (Theorem S8.1 for the nonlinear posterior update, Theorem S8.2 for the sequential information bound) and declares the theorem/computation boundary: every reported "optimal" Bayesian design is computed, not symbolically derived, and its validity rests on convergence diagnostics rather than a closed form. The computational layer itself — the nested Monte Carlo estimator with its variance-reduction hierarchy, the adaptive schedule and its stopping rules, and the mandatory robustness axes — is fully specified in the Supplementary Material (Section S5); none of it is executed in this paper.

The development is organized around four objects: the model $M \in \{1, \ldots, K\}$, the within-model parameter ϑ_M, the design d (the safe input and observation schedule of Sections S6–S7), and the future data Y. No Gaussian or linear approximation is implied at the level of the objectives below; the linear-Gaussian formulae of Section S6 reappear only as the fast screening tier of the variance-reduction hierarchy.

S8.1. Bayesian model-discrimination utility

Let $p(M)$ be the prior over models, $p(\vartheta_M \mid M)$ the within-model parameter prior, and $p(Y \mid \vartheta_M, M, d)$ the observation likelihood of model M at design d. The exact expected mutual information between model identity and future data is

$$U_{\mathrm{MI}}(d) = I(M; Y \mid d) = \mathbb{E}\left[\log \frac{p(Y \mid M, d)}{\sum_{j=1}^{K} p(M_j)\, p(Y \mid M_j, d)}\right], \tag{S42}$$

where the marginal likelihood of model j integrates out its parameters,

$$p(Y \mid M_j, d) = \int p(Y \mid \vartheta_{M_j}, M_j, d)\, p(\vartheta_{M_j} \mid M_j)\, d\vartheta_{M_j}. \tag{S43}$$

Equations (S42)–(S43) are the exact Bayesian model-discrimination objective; the linear-Gaussian quadratic (S18) is recovered only when the likelihood is Gaussian in the observable and the marginal (S43) is replaced by its Laplace approximation. A joint model-plus-parameter utility exists and is reported separately from (S42) whenever parameter recovery is also a goal; its definition and the reasons the two objectives must not be merged into a single scalar are given in the Supplementary Material (Section S5).

S8.2. Robust alternative

For model pairs with parameter sets Θ_i, Θ_j, a more adversarial objective protects against alternatives that tune themselves to mimic the focal model:

$$U_{\min}(d) = \min_{i<j} \inf_{\substack{\theta_i \in \Theta_i \\ \theta_j \in \Theta_j}} D_{\mathrm{KL}}\big(p_i(Y \mid \theta_i, d) \,\big\|\, p_j(Y \mid \theta_j, d)\big). \tag{S44}$$

The minimax objective (S44) collapses to zero whenever a pair of parameter sets admits a tuning that realizes likelihood overlap below the noise floor — which, by the finite-horizon impossibility result of Corollary S5.6, is exactly the regime in which an unrestricted latent class can mimic the fractional model. The complexity budget m imposed in Section S5 removes this unrestricted-dimension degeneracy, but it does *not* by itself restore positivity of (S44): the robust separation can still vanish at boundary or degenerate parameter values — $\alpha \to 1$ collapsing onto the ODE model, $\tau \to 0$ collapsing onto the ODE model, zero latent coupling, or any parameter configuration producing identical sampled outputs. Positivity of (S44) requires, in addition, compact parameter sets and an explicit exclusion of such intersections (or a certified positive minimum distance under the chosen design). Where these exclusions are not enforced, the honest statement is that (S44) may attain zero even with bounded latent dimension.

S8.3. Theorem T18 — posterior update under nonlinear Gaussian likelihood

Equation (S42) defines the design utility, but the schedule that drives it is sequential: after each batch of data, the posterior over models and parameters updates, and the next design is chosen adaptively. The next theorem characterizes the posterior update under a general nonlinear observation map with additive Gaussian noise, which is the regime that bridges the linearized layer of Section S6 and the fully nonlinear layer demanded by the Bayesian objective.

Theorem S8.1 (Nonlinear posterior update under Gaussian noise)**.** *Let model M induce an observation of the form*

$$Y = \mu_M(\vartheta_M; d) + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, R), \tag{S45}$$

where μ_M is a (possibly nonlinear, differentiable) mean map from the parameter and the design to the observation space, and $R \succ 0$ is the noise covariance. Then the joint posterior over model identity and parameter, after data $Y_{1:n}$ collected under designs $d_{1:n}$, factorizes as

$$p(M, \vartheta_M \mid Y_{1:n}) \propto p(M)\, p(\vartheta_M \mid M) \prod_{k=1}^{n} \mathcal{N}\big(Y_k;\, \mu_M(\vartheta_M; d_k),\, R\big), \tag{S46}$$

and the model marginal posterior is

$$p(M \mid Y_{1:n}) = \frac{p(M)\, p(Y_{1:n} \mid M, d_{1:n})}{\sum_j p(M_j)\, p(Y_{1:n} \mid M_j, d_{1:n})}, \tag{S47}$$

with $p(Y_{1:n} \mid M, d_{1:n}) = \int \prod_k \mathcal{N}(Y_k; \mu_M(\vartheta_M; d_k), R)\, p(\vartheta_M \mid M) d\vartheta_M$. In the special linear-Gaussian conjugate case, where each $\mu_M(\vartheta_M; d_k) = H_{M,k}\vartheta_M + \mu^0_{M,k}$ is affine in the parameter and the prior $p(\vartheta_M \mid M)$ is Gaussian, the within-model marginal integrates exactly to a Gaussian predictive density, and the operator governing the leading-order model separation is the predictive-mean difference operator $Q = \Delta H^* R^{-1} \Delta H$ of Theorem S6.1. For a genuinely nonlinear mean map μ_M, the likelihood is Gaussian only conditionally on the parameter, the product likelihood is generically not Gaussian in ϑ_M, and the integral above is computed, not evaluated.

Proof idea. Bayes' rule and independence of the additive noise across steps give the product form; the conjugate case integrates exactly. The complete proof is given in the Supplementary Material (Section S5). □

S8.4. Safe Bayesian design

The Bayesian objective (S42) is maximized subject to a safety constraint that extends the deterministic certificate of Section S7 to the case where the parameters — and hence the safe region and the Allee threshold A — are uncertain. Two interpretations are distinguished and the paper reports which is in force.

The *chance* constraint averages safety over the current posterior:

$$\Pr\big(z(t) \in \mathcal{S}\ \ \forall t \in [0, T] \mid d, Y_{1:n}\big) \geq 1 - \delta_s, \tag{S48}$$

where the probability is taken jointly over the current model posterior, parameter posterior, and observation noise. The *robust* constraint, more conservative, requires safety for every parameter in a credible set Θ_{adm}:

$$z(t; \vartheta) \in \mathcal{S}\ \ \forall t \in [0, T],\ \ \forall \vartheta \in \Theta_{\text{adm}}. \tag{S49}$$

The deterministic rectangle inequalities of Section S7 reappear here as the worst-case check on Θ_{adm} via interval arithmetic over the Allee threshold A, the predator growth constants, and the intervention amplitudes. The chance formulation (S48) trades a small bound δ_s of posterior mass for far larger admissible regions and thus for higher information per experiment; the robust formulation (S49) provides a certificate that no admitted parameter triggers collapse, at the cost of information. The adaptive termination of Section S7 extends to the Bayesian layer: a trial halts automatically when the posterior predictive probability of crossing x_L exceeds a predeclared level.

S8.5. Theorem T19 — Bayesian sequential information bound

The adaptive schedule maximizes U_{MI} over designs drawn from a safe admissible set $\mathcal{D}_{\text{safe}}(Y_{1:n})$ that incorporates the chance constraint (S48) or, where required, the robust constraint (S49). The next theorem bounds the expected information gain of any single future experiment by the prior-posterior separation, which is the quantity the adaptive schedule actually optimizes.

Theorem S8.2 (Bayesian sequential information identity and bound)**.** *Let $p(M \mid Y_{1:n})$ be the model posterior after data $Y_{1:n}$ collected under designs $d_{1:n}$, and let d_{n+1} be any admissible next design. Then the one-step expected information gain admits the expected-KL representation*

$$I\big(M; Y_{n+1} \mid Y_{1:n}, d_{n+1}\big) = \mathbb{E}_{Y_{n+1}|Y_{1:n}, d_{n+1}}\left[D_{\text{KL}}\big(p(M \mid Y_{1:n}, Y_{n+1}, d_{n+1}) \,\big\|\, p(M \mid Y_{1:n})\big)\right], \tag{S50}$$

and satisfies

$$0 \le I(M; Y_{n+1} \mid Y_{1:n}, d_{n+1}) \le H(M \mid Y_{1:n}). \tag{S51}$$

Equality in the upper bound holds if and only if the future observation determines the model identity almost surely conditional on the current data and design, i.e. $H(M \mid Y_{1:n}, Y_{n+1}, d_{n+1}) = 0$.

Proof idea. The chain rule for mutual information, expansion of the conditional mutual information as an expectation of log-ratios, and nonnegativity of conditional entropy. The complete proof, together with the tower-property caveat (the identity is an expectation of divergences between *realized* posteriors — replacing it by the divergence of the *expected* posterior yields identically zero) and the honest status of greedy one-step scheduling (myopic, with no certified non-myopic optimality), is given in the Supplementary Material (Section S5). □

S8.6. Computational layer and theorem/computation boundary

Outside the linear-Gaussian regime, the objectives (S42) and (S50) have no closed form. The Supplementary Material (Section S5) specifies the complete computational layer: a nested Monte Carlo estimator with common random numbers; a four-tier variance-reduction hierarchy (exact linear-Gaussian screening, Laplace, importance sampling / sequential Monte Carlo, calibrated neural bounds) in which each tier is admissible only after recovering the tier below on a case where the latter is exact; the adaptive schedule with its stopping rules; and eight mandatory robustness axes against which every reported design must be re-evaluated. The analytical content of this section is the two theorems and the boundary they draw; the remainder is computation, and the paper is explicit about that boundary:

> The nonlinear Bayesian optimum is computed, not symbolically derived. Its validity is established by convergence diagnostics, independent re-simulation, utility confidence intervals, and recovery on analytically solvable benchmarks.

Figure S12 summarizes this future operational layer as a loop: current belief ⟶ safe admissible designs ⟶ expected utility via Theorem S8.2 ⟶ execution ⟶ posterior update via Theorem S8.1 ⟶ stopping rule. The loop is specified here; it is not executed in the present benchmark, whose decision rule is BIC (the main-paper benchmark).

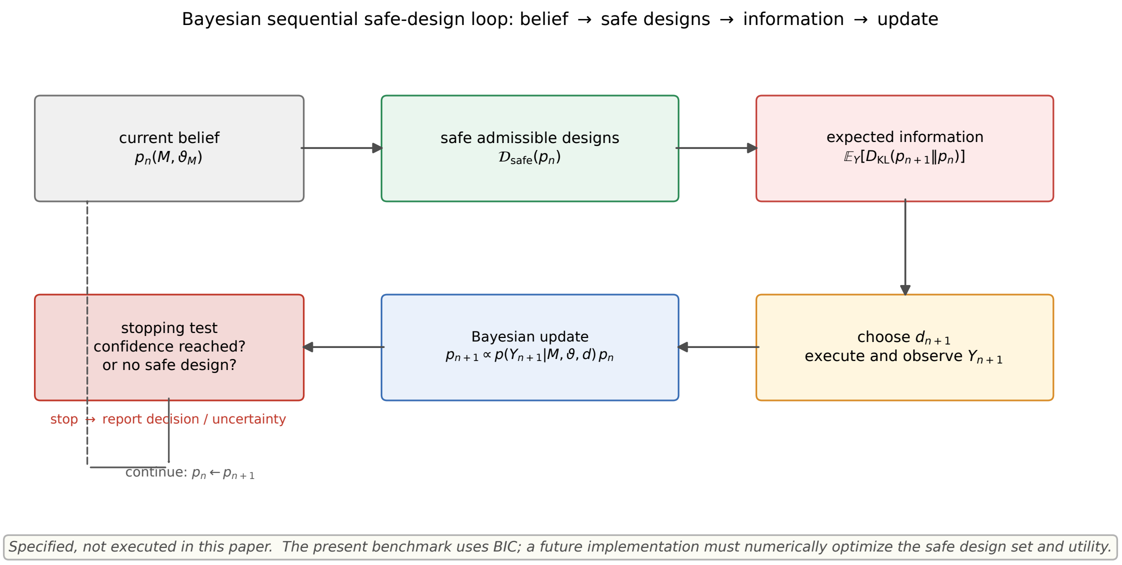

Figure S12. The Bayesian sequential safe-design loop specified in this section: current belief, posterior-conditioned safe admissible designs, expected utility via the expected-KL identity of Theorem S8.2, execution and posterior update via Theorem S8.1, and an explicit stop/continue decision. The loop is pinned down analytically and left to future work as an executed layer; the present benchmark uses BIC (the main-paper benchmark).

S9. Fractional-delay numerical formulation and validation

The main paper introduces the combined fractional-delay model

$$\tau_0^{\alpha-1}{}^C D_t^\alpha x(t) = P(x(t);A) - \frac{ax(t)y(t)}{1+hx(t)} + u(t),$$
$$\tau_0^{\alpha-1}{}^C D_t^\alpha y(t) = e\frac{ax(t-\tau)y(t)}{1+hx(t-\tau)} - my(t).$$

A constant equilibrium history is prescribed on $[-\tau, 0]$. The PECE implementation uses the same fractional Adams–Bashforth–Moulton weights as the pure Caputo solver and linearly interpolates the delayed prey history. For every reported $\tau > 0$, the delay exceeds the numerical time step, so the delayed state required by the corrector is already available.

Two numerical limits are checked independently. First, at $\tau = 0$ the implementation calls the validated pure Caputo solver exactly. Second, at $\alpha = 1$ it converges to the independent interpolated-history RK4 DDE implementation. At $A = 0.25$, $\tau = 0.35$, $T = 6$, the maximum absolute discrepancies are

N	200	400	800
$\max \lvert z_{FD} - z_{DDE} \rvert$	6.75×10^{-6}	1.68×10^{-6}	4.20×10^{-7}

For the representative cell $(\alpha, \tau) = (0.85, 0.35)$, the max difference from the $N = 800$ trajectory decreases from 2.18×10^{-3} at $N = 200$ to 8.05×10^{-4} at $N = 400$.

The targeted design study uses $A = 0.25$, $T = 12$, $\alpha \in \{0.75, 0.85, 0.95\}$, $\tau \in \{0.15, 0.30, 0.45\}$, and the same six peak-limited inputs as the main benchmark. The resulting mean minimum trajectory distance to the pure Caputo and pure DDE limits, together with numerical Allee-crossing fractions, is

Design	mean minimum bridge distance	Allee-crossing fraction
multisine	0.1332	0.333
PRBS	0.0909	1.000
sinusoid	0.0894	1.000
chirp	0.0861	1.000
pulse	0.0627	0.000
multiscale	0.0338	0.000

These quantities are descriptive numerical diagnostics, not certificates.

S10. Fractional solver validation

This appendix collects numerical-method validation that supports the benchmark of the main-paper benchmark but is not part of its headline results.

The Caputo solver used throughout is the fractional Adams–Bashforth–Moulton PECE scheme of `benchmark/core.py`. Its convergence is validated against the exact Mittag–Leffler solution of the scalar test problem $D^\alpha y = \lambda y$, $y(0) = 1$, whose solution is $y(t) = E_\alpha(\lambda t^\alpha)$. Figure S13 shows the maximum absolute error as a function of the number of time steps on the benchmark grid; the error decreases under refinement and reaches 2.5×10^{-4} at the production resolution, as reported in the main-paper benchmark.

References

1. Anthony C. Atkinson and Valerii V. Fedorov. The design of experiments for discriminating between two rival models. *Biometrika*, 62(1):57–70, 1975.

2. Anthony C. Atkinson and Valerii V. Fedorov. Optimal design: Experiments for discriminating between several models. *Biometrika*, 62(2):289–303, 1975.

3. Luděk Berec, Elena Angulo, and Franck Courchamp. Multiple allee effects and population management. *Trends in Ecology & Evolution*, 22(4):185–191, 2007.

4. Gregory Beylkin and Lucas Monzón. On approximation of functions by exponential sums. *Applied and Computational Harmonic Analysis*, 19(1):17–48, 2005.

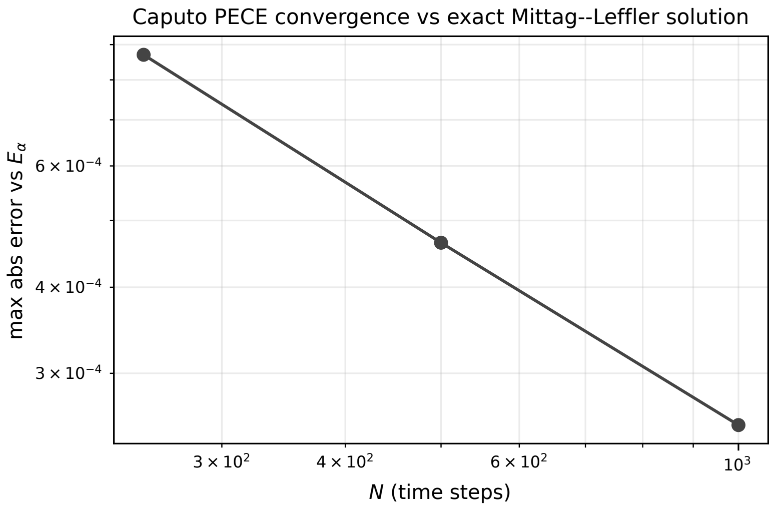

Figure S13. Caputo PECE solver convergence against the exact Mittag–Leffler solution: maximum absolute error versus the number of time steps, on a log–log scale. The monotone decrease under refinement is the numerical gate behind every nonlinear cell of the main-paper benchmark.

5. Gregory Beylkin and Lucas Monzón. Approximation by exponential sums revisited. *Applied and Computational Harmonic Analysis*, 28(2):131–149, 2010.

6. Kathryn Chaloner and Isabella Verdinelli. Bayesian experimental design: A review. *Statistical Science*, 10(3):273–304, 1995.

7. Herman Chernoff. A measure of asymptotic efficiency for tests of a hypothesis based on the sum of observations. *The Annals of Mathematical Statistics*, 23(4):493–507, 1952.

8. Nguyen Dinh Cong, Thai Son Doan, Stefan Siegmund, and Hoang The Tuan. Linearized asymptotic stability for fractional differential equations. *Electronic Journal of Qualitative Theory of Differential Equations*, 2016(39):1–13, 2016.

9. Franck Courchamp, Tim Clutton-Brock, and Bryan Grenfell. Inverse density dependence and the allee effect. *Trends in Ecology & Evolution*, 14(10):405–410, 1999.

10. Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley, 2 edition, 2006.

11. Holger Dette, Viatcheslav B. Melas, and Petr Shpilev. Robust T-optimal discriminating designs. *The Annals of Statistics*, 41(4):1693–1715, 2013.

12. Kai Diethelm. *The Analysis of Fractional Differential Equations: An Application-Oriented Exposition Using Differential Operators of Caputo Type*, volume 2004 of *Lecture Notes in Mathematics*. Springer, 2010.

13. Robert J. Flassig and Kai Sundmacher. Optimal design of stimulus experiments for robust discrimination of biochemical reaction networks. *Bioinformatics*, 28(23):3089–3096, 2012.

14. Jack K. Hale and Sjoerd M. Verduyn Lunel. *Introduction to Functional Differential Equations*, volume 99 of *Applied Mathematical Sciences*. Springer, 1993.

15. Xun Huan and Youssef M. Marzouk. Simulation-based optimal bayesian experimental design for nonlinear systems. *Journal of Computational Physics*, 232(1):288–317, 2013.

16. Shidong Jiang, Jiwei Zhang, Qian Zhang, and Zhimin Zhang. Fast evaluation of the caputo fractional derivative and its applications to fractional diffusion equations. *Communications in Computational Physics*, 21(3):650–678, 2017.

17. Thomas Kailath. *Linear Systems*. Prentice Hall, 1980.

18. Steven M. Kay. *Fundamentals of Statistical Signal Processing, Volume II: Detection Theory*. Prentice Hall, 1998.

19. Anatoly A. Kilbas, Hari M. Srivastava, and Juan J. Trujillo. *Theory and Applications of Fractional Differential Equations*, volume 204 of *North-Holland Mathematics Studies*. Elsevier, 2006.

20. Clemens Kreutz and Jens Timmer. Systems biology: Experimental design. *The FEBS Journal*, 276(4):923–942, 2009.

21. Solomon Kullback and Richard A. Leibler. On information and sufficiency. *The Annals of Mathematical Statistics*, 22(1):79–86, 1951.

22. Changpin Li and Yujiang Ma. Fractional dynamical system and its linearization theorem. *Nonlinear Dynamics*, 71(4):621–633, 2013.

23. Yan Li, YangQuan Chen, and Igor Podlubny. Mittag–leffler stability of fractional order nonlinear dynamic systems. *Automatica*, 45(8):1965–1969, 2009.

24. Christian Lubich. Discretized fractional calculus. *SIAM Journal on Mathematical Analysis*, 17(3):704–719, 1986.

25. Maximiliano Lucius. Computer-assisted stability and extinction certificates for a caputo predator–prey system with a strong prey allee effect. Companion validated manuscript, draft dated 2026-07-30; published as Zenodo preprint DOI 10.5281/zenodo.21809908 (companion project fractional-memory-identifiability), 2026.

26. Jesús López-Fidalgo, Chiara Tommasi, and Paula C. Trandafir. An optimal experimental design criterion for discriminating between non-normal models. *Journal of the Royal Statistical Society: Series B*, 69(2):231–242, 2007.

27. Denis Matignon. Stability results for fractional differential equations with applications to control processing. In *Computational Engineering in Systems Applications*, volume 2, pages 963–968, Lille, France, 1996.

28. William McLean. Numerical evaluation of mittag–leffler functions. *Calcolo*, 58(1):7, 2021.

29. Wim Michiels and Silviu-Iulian Niculescu. *Stability, Control, and Computation for Time-Delay Systems: An Eigenvalue-Based Approach*. SIAM, 2 edition, 2014.

30. Gérard Montseny. Diffusive representation of pseudo-differential time-operators. *ESAIM: Proceedings*, 5:159–175, 1998.

31. Igor Podlubny. *Fractional Differential Equations*, volume 198 of *Mathematics in Science and Engineering*. Academic Press, 1999.

32. Tom Rainforth, Adam Foster, Desi R. Ivanova, and Freddie Bickford Smith. Modern bayesian experimental design. *Statistical Science*, 39(1), 2024.

33. Elizabeth G. Ryan, Christopher C. Drovandi, James M. McGree, and Anthony N. Pettitt. A review of modern computational algorithms for bayesian optimal design. *International Statistical Review*, 84(1):128–154, 2016.

34. Achim Schädle, María López-Fernández, and Christian Lubich. Fast and oblivious convolution quadrature. *SIAM Journal on Scientific Computing*, 28(2):421–438, 2006.

35. Dominik Skanda and Dirk Lebiedz. An optimal experimental design approach to model discrimination in dynamic biochemical systems. *Bioinformatics*, 26(7):939–945, 2010.

36. Philip A. Stephens, William J. Sutherland, and Robert P. Freckleton. What is the allee effect? *Oikos*, 87(1):185–190, 1999.

37. Harry L. Van Trees and Kristine L. Bell. *Detection, Estimation, and Modulation Theory, Part I*. Wiley, 2 edition, 2013.

38. Haiping Ye, Jianming Gao, and Yongsheng Ding. A generalized grönwall inequality and its application to a fractional differential equation. *Journal of Mathematical Analysis and Applications*, 328(2):1075–1081, 2007.